JOSHUA T. VOGELSTEIN PH.D.

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ASSOCIATE PROFESSOR, BIOMEDICAL ENGINEERING, JOHNS HOPKINS UNIVERSITY

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PERSONAL INFORMATION

PRIMARY APPOINTMENT

02/22 – **Associate Professor**, *Department of Biomedical Engineering*, *JHU, Baltimore, MD, USA*

08/14 – 02/22 **Associate Professor**, *Department of Biomedical Engineering*, *JHU, Baltimore, MD, USA*

JOINT APPOINTMENTS

09/19 – **Joint Appointment**, *Department of Biostatistics*, *JHU, Baltimore, MD, USA*

08/15 – **Joint Appointment**, *Department of Applied Mathematics and Statistics*, *JHU, Baltimore, MD, USA*

08/14 – **Joint Appointment**, *Department of Neuroscience*, *JHU, Baltimore, MD, USA*

08/14 – **Joint Appointment**, *Department of Computer Science*, *JHU, Baltimore, MD, USA*

INSTITUTIONAL AND CENTER APPOINTMENTS

08/15 – 05/26 **Steering Committee**, *Kavli Neuroscience Discovery Institute (KNDI)*, *Baltimore, MD, USA*

08/14 – **Core Faculty**, *Institute for Computational Medicine*, *JHU, Baltimore, MD, USA*

08/14 – **Core Faculty**, *Center for Imaging Science*, *JHU, Baltimore, MD, USA*

08/14 – **Assistant Research Faculty**, *Human Language Technology Center of Excellence*, *JHU, Baltimore, MD, USA*

10/12 – **Affiliated Faculty**, *Institute for Data Intensive Engineering and Sciences*, *JHU, Baltimore, MD, USA*

10/25 – **Faculty**, *Mathematical Institute for Data Science (MINDS)*, *JHU, Baltimore, MD, USA*

02/26 – **Faculty**, *Data Science and AI Institute (DSAI)*, *JHU, Baltimore, MD, USA*

EDUCATION & TRAINING

2003 – 2009 **Ph.D in Neuroscience**, *JHU School of Medicine*, *Advisor: Eric Young*
Thesis: OOPSI: a family of optical spike inference algorithms for inferring neural connectivity from population calcium imaging

2009 – 2009 **M.S. in Applied Mathematics & Statistics**, *JHU*

1998 – 2002 **B.A. in Biomedical Engineering**, *Washington University, St. Louis*

2022 *Foundations in Somatic Abolitionism for White Bodies (14 Hours)*

2022 *Sacred Sons Leadership Training Level 1*

2022 *Creatorhood Initiate Training - Phase 1*

2021 *JHU SafeZone training*

ACADEMIC EXPERIENCE

08/18 – **Director of Biomedical Data Science Focus Area**, *Department of Biomedical Engineering*, *JHU, Baltimore, MD, USA*

05/16 – **Visiting Scientist**, *Howard Hughes Medical Institute*, *Janelia Research Campus, Ashburn, VA, USA*

10/12 – 08/14 **Endeavor Scientist**, *Child Mind Institute*, *New York, NY, USA*

08/12 – 08/14 **Affiliated Faculty**, *Kenan Institute for Ethics*, *Duke University, Durham, NC, USA*

08/12 – 08/14 **Adjunct Faculty**, *Department of Computer Science*, *JHU, Baltimore, MD, USA*

12/09 – 01/11 **Post-Doctoral Fellow**, *Department of Applied Mathematics and Statistics*, *JHU, Baltimore, MD, USA*
Supervised by Carey E. Priebe, **Research** Statistics of populations of networks.

06/01 – 09/01 **Research Assistant**, *Prof. Randy O'Reilly, Dept. of Psychology*, *University of Colorado, Denver, CO, USA*

06/00 – 09/00 **Clinical Engineer**, *Johns Hopkins Hospital*, *JHU, Baltimore, MD, USA*

06/99 – 08/99 **Research Assistant under Dr. Jeffrey Williams**, *Dept. of Neurosurgery, Johns Hopkins Hospital*, *Baltimore, MD, USA*

06/98 – 08/98 **Research Assistant under Professor Kathy Cho**, *Dept. of Pathology, Johns Hopkins School of Medicine*, *Baltimore, MD, USA*

PUBLISHED RESEARCH

PUBLISHED PEER-REVIEWED RESEARCH ARTICLES

Note: CV author in bold; trainees are underlined.

126 papers; 19,457 total citations; top 10 cited 8,722 times; H-index 60; 13 first, 33 last, 78 middle authorships (Google Scholar, as of September 22, 2026).

[126] Cencheng Shen, Jesús Arroyo, Junhao Xiong, and **Joshua T. Vogelstein**. "Community correlations and testing independence between binary graphs." *Applied Network Science*, (Apr 2026). https://doi.org/10.1007/s41109-026-00791-6

[125] Tingshan Liu, Jayanta Dey, Beiya Xu, Eric W. Bridgeford, Samuel Alldritt, Karl-Heinz Nenning, Kyoungseob Byeon, Ting Xu, and **Joshua T. Vogelstein**. "Statistically valid explainable black-box machine learning: applications in sex classification across species using brain imaging." *PLOS ONE*, (Apr 2026). https://doi.org/10.1371/journal.pone.0346575

[124] Kseniia Konishcheva, Bennett Leventhal, Maki Koyama, Sambit Panda, **Joshua T. Vogelstein**, Michael Milham, Ariel Lindner, and Arno Klein. "Accurate and efficient data-driven psychiatric assessment using machine learning." *BMC Medical Informatics and Decision Making*, (Jan 2026). https://doi.org/10.1186/s12911-025-03329-5

[123] Konrad Kording, **Joshua T. Vogelstein**, Pratik Chaudhari, and Timothy Verstynen. "Toward a science of prospective learning." *Neuron*, (Nov 2025). https://www.cell.com/neuron/fulltext/S0896-6273(25)00857-8?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0896627325008578%3Fshowall%3Dtrue

[122] Samuel D. Curtis, Sambit Panda, Adam Li, Haoyin Xu, Yuxin Bai, Itsuki Ogihara, Eliza O'Reilly, Yuxuan Wang, Lisa Dobbyn, Maria Popoli, Janine Ptak, Nadine Nehme, Natalie Silliman, Jeanne Tie, Peter Gibbs, Lan T. Ho-Pham, Bich N.H. Tran, Thach S. Tran, Tuan V. Nguyen, Ehsan Irajizad, Michael Goggins, Christopher L Wolfgang, Tian-Li Wang, Ie-Ming Shih, Amanda Fader, Anne Marie Lennon Lennon, Ralph H. Hruban, Chetan Bettegowda, Lucy Gilbert, Kenneth W. Kinzler, Nickolas Papadopoulos, Bert Vogelstein, **Joshua T. Vogelstein**, and Christopher Douville. "Minimizing and quantifying uncertainty in AI-informed decisions: Applications in medicine." *PNAS*, (Aug 2025). https://www.pnas.org/doi/10.1073/pnas.2424203122

[121] Samuel D Curtis, Tingshan Liu, Yuxin Bai, Yuxuan Wang, Sambit Panda, Adam Li, Haoyin Xu, Eliza O'Reilly, Lisa Dobbyn, Maria Popoli, Janine Ptak, Natalie Silliman, Chris Thoburn, Jeanne Tie, Peter Gibbs, Lan T Ho-Pham, Bich N.H. Tran, Thach S Tran, Tuan V Nguyen, Maximilian F Konig, Michelle Petri, Antony Rosen, Christopher A Mecoli, Ami A Shah, Frits Mulder, Nick van Es, PLATO-VTE Study Group, Chetan Bettegowda, Kenneth W Kinzler, Nickolas Papadopoulos, **Joshua T Vogelstein**, Bert Vogelstein, and Christopher Douville. "Fragmentation signatures in cancer patients resemble those of patients with vascular or autoimmune diseases." *PNAS*, (Aug 2025). https://www.pnas.org/doi/10.1073/pnas.2426890122

[120] Jayanta Dey, **Joshua T Vogelstein**, Hayden S Helm, Will LeVine, Ronak D Mehta, Tyler M Tomita, Haoyin Xu, Ali Geisa, Qingyang Wang, Gido M van de Ven, and others. "Simple Lifelong Learning Machines." *Transactions on Pattern Analysis and Machine Intelligence*, (Jul 2025). https://ieeexplore.ieee.org/document/11108956/authors#authors

[119] Juan M Lavista Ferres, Elliot K Fishman, Ed Catmull, Bert Vogelstein, and **Joshua T. Vogelstein**. "Translating Artificial Intelligence Breakthroughs into Cancer Diagnostic Breakthroughs." *Cancer Discovery*, (Jul 2025). https://tnq-aacr.proofcentral.com/en-us/index.html?token=2da22a4j07ffcae42d13503d324962

[118] Eric W. Bridgeford, Jaewon Chung, Robert J. Anderson, Ali Mahzarnia, Jacques A. Stout, Hae Sol Moon, Zay Yar Han, **Joshua T. Vogelstein**, and Alexandra Badea. "Network Biomarkers of Alzheimer's Disease Risk Derived from Joint Volume and Texture Covariance Patterns in Mouse Models." *Imaging Neuroscience*, (Jun 2025). https://pubmed.ncbi.nlm.nih.gov/39975084/

[117] Jayanta Dey, Haoyin Xu, Ashwin De Silva, and **Joshua T Vogelstein**. "Simple Calibration via Geodesic Kernels." *Transactions of Machine Learning Research*, (Jan 2025). https://arxiv.org/abs/2201.13001

[116] Sambit Panda, Cencheng Shen, Ronan Perry, Jelle Zorn, Antoine Lutz, Carey E. Priebe, and **Joshua T. Vogelstein**. "Universally Consistent K-sample Tests via Dependence Measures." *Statistics and Probability Letters*, (Jan 2025). https://doi.org/10.1016/j.spl.2024.110278

[115] Sambit Panda, Shen Cencheng, Ronan Perry, Jelle Zorn, Antoine Lutz, Carey E. Priebe, and **Joshua T. Vogelstein**. "Universally consistent K-sample tests via dependence measures." *Statistics and probability Letters*, (Jan 2025). https://www.sciencedirect.com/science/article/pii/S0167715224002475

[114] Eric W. Bridgeford, Michael Powell, Gregory Kiar, Stephanie Noble, Jaewon Chung, Sambit Panda, Ross Lawrence, Ting Xu, Michael Milham, Brian Caffo, and **Joshua T. Vogelstein**. "When no answer is better than a wrong answer: a causal perspective on batch effects." *Imaging Neuroscience*, (Dec 2024). https://doi.org/10.1101/2021.09.03.458920

[113] Ting Xu, Gregory Kiar, Xi-Nian Zuo, **Joshua T. Vogelstein**, and Michael P. Milham. "Challenges in Measuring Individual Differences of Brain Function." *Imaging Neuroscience*, (Dec 2024). https://direct.mit.edu/imag/article/doi/10.1162/imag_a_00430/125636/Perspective-Challenges-in-Measuring-Individual

[112] Guodong Chen, Jesús Arroyo, Avanti Athreya, Joshua Cape, **Joshua T. Vogelstein**, Youngser Park, Chris White, Jonathan Larson, Weiwei Yang, and Carey E. Priebe. "Multiple Network Embedding for Anomaly Detection in Time Series of Graphs." *Computational Statistics and Data Analysis*, (Sep 2024). https://arxiv.org/abs/2008.10055

[111] Shen Cencheng, Shangi Wang, Alexandra Badea, Carey E. Priebe, and **Joshua T. Vogelstein**. "Discovering the signal subgraph: An iterative screening approach on graphs." *Pattern Recognition Letters*, (Jun 2024). https://www.sciencedirect.com/science/article/pii/S0167865524001818

[110] Xinhui Li, Nathalia Bianchini Esper, Lei Ai, Steve Giavasis, Hecheng Jin, Eric Feczko, Ting Xu, Jon Clucas, Alexandre Franco, Anibal Sólon Heinsfeld, Azeez Adebimpe, **Joshua T. Vogelstein**, Chao-Gan Yan, Oscar Esteban, Russell A. Poldrack, Cameron Craddock, Damien Fair, Theodore Satterthwaite, Gregory Kiar, and Michael P. Milham. "Moving Beyond Processing and Analysis-Related Variation in Neuroscience." *Nature Human Behaviour*, (May 2024). https://doi.org/10.1101/2021.12.01.470790

[109] Shen Cencheng, Chung Jaewon, Mehta Ronak, Xu Ting, and **Joshua T. Vogelstein**. "Independence Testing for Temporal Data." *Transactions on Machine Learning Research*, (May 2024). https://arxiv.org/abs/1908.06486

[108] Hayden S. Helm, Ashwin De Silva, **Joshua T and Vogelstein**, Carey E. Priebe, and Weiwei Yang. "Approximately Optimal Domain Adaptation with Fisher's Linear Discriminant." *Mathmatics*, (Feb 2024). https://arxiv.org/abs/2302.14186

[107] Zeyi Wang, Eric Bridgeford, Shangsi Wang, **Joshua T. Vogelstein**, and Brian Caffo. "Statistical Analysis of Data Repeatability Measures." *International Statistical Review*, (2024). https://arxiv.org/abs/2005.11911

[106] Tiona Zuzul, Emily Cox Pahnke, Jonathan Larson, Patrick Bourke, Nicholas Caurvina, Neha Parikh Shah, Fereshteh Amini, Jeffrey Weston, Youngser Park, **Joshua Vogelstein**, Christopher White, and Carey E. Priebe. "Dynamic Silos: Increased Modularity and Decreased Stability in Intraorganizational Communication Networks During the COVID-19 Pandemic." *Management Science*, (2024). https://doi.org/10.1287/mnsc.2022.02797

[105] Meghana Madhyastha, Tamas Budavari, Vladmir Braverman, **Joshua Vogelstein**, and Randal Burns. "T-Rex (Tree-Rectangles): Reformulating Decision Tree Traversal as Hyperrectangle Enclosure." *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, (2024). https://doi.org/10.1109/ICDE60146.2024.00145

[104] Thomas L Athey, Matthew A Wright, Marija Pavlovic, Vikram Chandrashekhar, Karl Deisseroth, Michael I Miller, and **Joshua T Vogelstein**. "BrainLine: An Open Pipeline for Connectivity Analysis of Heterogeneous Whole-Brain Fluorescence Volumes." *Neuroinformatics*, (Oct 2023). https://doi.org/10.1007/s12021-023-09638-2

[103] Ruoxuan Xiong, Allison Koenecke, Michael Powell, Zhu Shen, **Joshua T Vogelstein**, and Susan Athey. "Federated Causal Inference in Heterogeneous Observational Data." *Statistics in Medicine*, (Aug 2023).

[102] Michael Winding, Benjamin D Pedigo, Christopher L Barnes, Heather G Patsolic, Youngser Park, Tom Kazimiers, Akira Fushiki, Ingrid V Andrade, Avinash Khandelwal, Javier Valdes-Aleman, Feng Li, Nadine Randel, Elizabeth Barsotti, Ana Correia, Richard D Fetter, Volker Hartenstein, Carey E Priebe, **Joshua T Vogelstein**, Albert Cardona, and Marta Zlatic. "The connectome of an insect brain." *Science*, (Mar 2023). https://www.science.org/doi/abs/10.1126/science.add9330

[101] Benjamin D Pedigo, Mike Powell, Eric W Bridgeford, Michael Winding, Carey E Priebe, and **Joshua T Vogelstein**. "Generative network modeling reveals quantitative definitions of bilateral symmetry exhibited by a whole insect brain connectome." *eLife Sciences Publications, Ltd*, (Mar 2023). https://elifesciences.org/articles/83739

[100] Ting Xu, JaeWook Cho, Gregory Kiar, Eric W Bridgeford, **Joshua T Vogelstein**, and Michael P Milham. "ReX: an integrative tool for quantifying and optimizing measurement reliability for the study of individual differences." *Nature Methods*, (2023). https://doi.org/10.1038/s41592-023-01901-3

[99] Benjamin D. Pedigo, Michael Winding, Carey E. Priebe, and **Joshua T. Vogelstein**. "Bisected graph matching improves automated pairing of bilaterally homologous neurons from connectomes." *Network Neuroscience*, (May 2022). https://www.biorxiv.org/content/10.1101/2022.05.19.492713

[98] Thomas L Athey, Daniel J Tward, Ulrich Mueller, **Vogelstein Joshua T**, and Michael I Miller. "Hidden Markov modeling for maximum probability neuron reconstruction." *Communications Biology*, (Apr 2022). https://doi.org/10.1038/s42003-022-03320-0

[97] Dhireesha Kudithipudi, Mario Aguilar-Simon, Jonathan Babb, Maxim Bazhenov, Douglas Blackiston, Josh Bongard, Andrew P. Brna, Suraj Chakravarthi Raja, Nick Cheney, Jeff Clune, Anurag Daram, Stefano Fusi, Peter Helfer, Leslie Kay, Nicholas Ketz, Zsolt Kira, Soheil Kolouri, Jeffrey L. Krichmar, Sam Kriegman, Michael Levin, Sandeep Madireddy, Santosh Manicka, Ali Marjaninejad, Bruce McNaughton, Risto Miikkulainen, Zaneta Navratilova, Tej Pandit, Alice Parker, Praveen K. Pilly, Sebastian Risi, Terrence J. Sejnowski, Andrea Soltoggio, Nicholas Soures, Andreas S. Tolias, Darío Urbina-Meléndez, Francisco J. Valero-Cuevas, Gido M. van de Ven, **Joshua T. Vogelstein**, Felix Wang, Ron Weiss, Angel Yanguas-Gil, Xinyun Zou, and Hava Siegelmann. "Biological underpinnings for lifelong learning machines." *Nature Machine Intelligence*, (Mar 2022). https://doi.org/10.1038/s42256-022-00452-0

[96] Shilong Li, Tomi Jun, Jonathan Tyler, Emilio Schadt, Yu-Han Kao, Zichen Wang, Maximilian F Konig, Chetan Bettegowda, **Joshua T Vogelstein**, Nickolas Papadopoulos, Ramon E Parsons, Rong Chen, Eric E Schadt, Li Li, and William K Oh. "Inpatient Administration of Alpha-1-Adrenergic Receptor Blocking Agents Reduces Mortality in Male COVID-19 Patients." *Front. Med.*, (Feb 2022). https://www.frontiersin.org/articles/10.3389/fmed.2021.637647/full

[95] Jean-Baptiste Poline, David N Kennedy, Friedrich T Sommer, Giorgio A Ascoli, David C Van Essen, Adam R Ferguson, Jeffrey S Grethe, Michael J Hawrylycz, Paul M Thompson, Russell A Poldrack, Satrajit S Ghosh, David B Keator, Thomas L Athey, **Joshua T Vogelstein**, Helen S Mayberg, and Maryann E Martone. "Is Neuroscience FAIR? A Call for Collaborative Standardisation of Neuroscience Data." *Neuroinformatics*, (Jan 2022). https://link.springer.com/article/10.1007/s12021-021-09557-0

[94] Thomas Hartung, Lena Smirnova, Itzy E. M. Pantoja, Akwasi Akwaboah, Dowlette-Mary A. E. Din, Cindy Berlinicke, J L. Boyd, Brian S. Caffo, Ben Cappiello, Tzahi Cohen-Karni, Lowry Curley, Ralph Etienne-Cummings, Raha Dastgheyb, David H. Gracias, Frederic Gilbert, Christa W. Habela, Fang Han, Tim Harris, Kathrin Herrmann, Eric J. Hill, Qi Huang, Rabih E. Jabbour, Erik C. Johnson, Brett J. Kagan, Caroline Krall, Andre Levchenko, Paul Locke, Alexandra Maertens, Monica Metea, Alysson R. Muotri, Rheinallt Parri, Barton L. Paulhamus, Jesse D. Plotkin, Paul Roach, July C. Romero, Jens C. Schwamborn, Fenna Sille, Alexander Szalay, Katya Tsaioun, Daniel Tornero, **Joshua T. Vogelstein**, Karl Wahlin, and Donald J. Zack. "The Baltimore Declaration toward the exploration of organoid intelligence." *Frontiers in Science*, (2022).

[93] Jaewon Chung, Bijan Varjavand, Jesús Arroyo-Relión, Anton Alyakin, Joshua Agterberg, Minh Tang, Carey E Priebe, and **Joshua T Vogelstein**. "Valid two-sample graph testing via optimal transport Procrustes and multiscale graph correlation with applications in connectomics." *Stat*, (2022).

[92] Mike Powell, Callahan Clark, Anton Alyakin, **Joshua T Vogelstein**, and Brian Hart. "Exploration of Residual Confounding in Analyses of Associations of Metformin Use and Outcomes in Adults With Type 2 Diabetes." *JAMA Network Open*, (2022). https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2798322

[91] Michael Powell, Allison Koenecke, James Byrd, Akihiko Nishimura, Maximilian Konig, Ruoxuan Xiong, Sadiqa Mahmood, Veraand Bettegowda, Chetan Mucaj, Liam Rose, Suzanne Tamang, Adam Sacarny, Brian Caffo, Susan Athey, Elizabeth Stuart, and **Joshua Vogelstein**. "Ten Rules for Conducting Retrospective Pharmacoepidemiological Analyses: Example COVID-19 Study." *Frontiers in Pharmacology*, (Jul 2021). https://www.frontiersin.org/articles/10.3389/fphar.2021.700776/full

[90] Vikram Chandrashekhar, Daniel J Tward, Devin Crowley, Ailey K Crow, Matthew A Wright, Brian Y Hsueh, Felicity Gore, Timothy A Machado, Audrey Branch, Jared S Rosenblum, Karl Deisseroth, and **Joshua T Vogelstein**. "CloudReg: automatic terabyte-scale cross-modal brain volume registration." *Nature Methods*, (Jul 2021). https://doi.org/10.1038/s41592-021-01218-z

[89] Thomas L Athey, Jacopo Teneggi, **Joshua T Vogelstein**, Daniel Tward, Ulrich Mueller, and Michael I Miller. "Fitting Splines to Axonal Arbors Quantifies Relationship between Branch Order and Geometry." *Frontiers in Neuroinformatics*, (Jun 2021). https://www.frontiersin.org/articles/10.3389/fninf.2021.704627/full? utm_source=Email_to_authors_ utm_medium=Email utm_content=T1_11.5e1_author utm_campaign=Email_publication field= journalName=Frontiers_in_Neuroinformatics id=704627

[88] Cencheng Shen, Sambit Panda, and **Joshua T. Vogelstein**. "The Chi-Square Test of Distance Correlation." *Journal of Computational and Graphical Statistics*, (Jun 2021). https://www.tandfonline.com/doi/full/10.1080/10618600.2021.1938585

[87] Allison Koenecke, Michael Powell, Ruoxuan Xiong, Zhu Shen, Nicole Fischer, Sakibul Huq, Adham M. Khalafallah, Marco Trevisan, Pr Sparen, Juan J Carrero, Akihiko Nishimura, Brian Caffo, Elizabeth A. Stuart, Renyuan Bai, Verena Staedtke, David L. Thomas, Nickolas Papadopoulos, Kenneth W. Kinzler, Bert Vogelstein, Shibin Zhou, Chetan Bettegowda, Maximilian F. Konig, Brett Mensh, **Joshua T. Vogelstein**, and Susan Athey. "Alpha-1 adrenergic receptor antagonists to prevent hyperinflammation and death from lower respiratory tract infection." *eLife*, (Jun 2021). https://elifesciences.org/articles/61700

[86] Ronan Perry, Gavin Mischler, Richard Guo, Theodore Lee, Alexander Chang, Arman Koul, Cameron Franz, Hugo Richard, Iain Carmichael, Pierre Ablin, Alexandre Gramfort, and **Joshua T. Vogelstein**. "mvlearn: Multiview Machine Learning in Python." *Journal of Machine Learning Research*, (May 2021). http://jmlr.org/papers/v22/20-1370.html

[85] **Joshua T. Vogelstein**, Eric W. Bridgeford, Minh Tang, Da Zheng, Christopher Douville, Randal Burns, and Mauro Maggioni. "Supervised dimensionality reduction for big data." *Nature Communications*, (May 2021).

[84] Shilong Li, Tomi Jun, Zichen Wang, Yu-Han Kao, Emilio Schadt, Maximilian F Konig, Chetan Bettegowda, **Joshua T Vogelstein**, Nickolas Papadopoulos, Ramon E Parsons, and others. "COVID-19 outcomes among hospitalized men with or without exposure to alpha-1-adrenergic receptor blocking agents." *Frontiers in Medicine*, (Apr 2021). https://www.medrxiv.org/content/10.1101/2021.04.08.21255148v1.full

[83] Shangsi Wang, Jesús Arroyo, **Joshua T Vogelstein**, and Carey E Priebe. "Joint Embedding of Graphs." *Transactions on Pattern Analysis and Machine Intelligence*, (Apr 2021). https://ieeexplore.ieee.org/document/8889404

[82] Liam Rose, Laura Graham, Allison Koenecke, Michael Powell, Ruoxuan Xiong, Zhu Shen, Brett Mench, Kenneth W Kinzler, Chetan Bettegowda, Bert Vogelstein, and others. "The association between Alpha-1 adrenergic receptor antagonists and in-hospital mortality from COVID-19." *Frontiers in Medicine*, (Mar 2021). https://www.frontiersin.org/articles/10.3389/fmed.2021.637647/full

[81] Michael P. Milham, **Joshua T. Vogelstein**, and Ting Xu. "Removing the Reliability Bottleneck in Functional Magnetic Resonance Imaging Research to Achieve Clinical Utility." *JAMA Psychiatry*, (Jan 2021). https://jamanetwork.com/journals/jamapsychiatry/article-abstract/2774875

[80] Jesús Arroyo, Avanti Athreya, Joshua Cape, Guodong Chen, Carey E. Priebe, and **Joshua T. Vogelstein**. "Inference for Multiple Heterogenous Networks with a Common Invariant Subspace." *Journal of Machine Learning Research*, (2021). http://jmlr.org/papers/v22/19-558.html

[79] Ross M Lawrence, Eric W Bridgeford, Patrick E Myers, Ganesh C Arvapalli, Sandhya C Ramachandran, Derek A Pisner, Paige F Frank, Allison D Lemmer, Aki Nikolaidis, and **Joshua T Vogelstein**. "Standardizing human brain parcellations." *Scientific data*, (2021). https://www.nature.com/articles/s41597-021-00849-3

[78] Eric W Bridgeford, Shangsi Wang, Zeyi Wang, Ting Xu, Cameron Craddock, Jayanta Dey, Gregory Kiar, William Gray-Roncal, Carlo Colantuoni, Christopher Douville, and others. "Eliminating accidental deviations to minimize generalization error and maximize replicability: Applications in connectomics and genomics." *PLoS computational biology*, (2021). https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1009279

[77] Vivek Gopalakrishnan, Jaewon Chung, Eric Bridgeford, Benjamin D. Pedigo, Jesús Arroyo, Lucy Upchurch, G. Allan Johnsom, Nian Wang, Youngser Park, Carey E. Priebe, and **Joshua T. Vogelstein**. "Multiscale Comparative Connectomics." *Imaging Neuroscience*, (Dec 2020). https://arxiv.org/abs/2011.14990

[76] Ting Xu, Karl-Heinz Nenning, Ernst Schwartz, Seok-Jun Hong, **Joshua T. Vogelstein**, Alexandros Goulas, Damien A. Fair, Charles E. Schroeder, Daniel S. Margulies, Jonny Smallwood, Michael P. Milham, and Georg Langs. "Cross-species functional alignment reveals evolutionary hierarchy within the connectome." *NeuroImage*, (Dec 2020). http://www.sciencedirect.com/science/article/pii/S1053811920308326

[75] Seok-Jun Hong, Ting Xu, Aki Nikolaidis, Jonathan Smallwood, Daniel S. Margulies, Boris Bernhardt, **Joshua T. Vogelstein**, and Michael P. Milham. "Toward a connectivity gradient-based framework for reproducible biomarker discovery." *NeuroImage*, (Dec 2020). http://www.sciencedirect.com/science/article/pii/S1053811920308089

[74] Karl-Heinz Nenning, Ting Xu, Ernst Schwartz, Jesus Arroyo, Adelheid Woehrer, Alexandre R. Franco, **Joshua T. Vogelstein**, Daniel S. Margulies, Hesheng Liu, Jonathan Smallwood, Michael P. Milham, and Georg Langs. "Joint embedding: A scalable alignment to compare individuals in a connectivity space." *NeuroImage*, (Nov 2020). http://www.sciencedirect.com/science/article/pii/S1053811920307187

[73] Jae Wook Chow, Annachiara Korchmaros, **Joshua T. Vogelstein**, Michael P. Milham, and Ting Xu. "Impact of concatenating fMRI data on reliability for functional connectomics." *Neuroimage*, (Nov 2020).

[72] Nian Wang, Robert J Anderson, David G Ashbrook, Vivek Gopalakrishnan, Youngser Park, Carey E Priebe, Yi Qi, **Joshua T Vogelstein**, Robert W Williams, and Allan G Johnson. "Variability and heritability of mouse brain structure: Microscopic MRI atlases and connectomes for diverse strains." *NeuroImage (Cover Story)*, (Nov 2020). http://www.sciencedirect.com/science/article/pii/S1053811920307606

[71] Melissa A Haendel, Christopher G Chute, Tellen D Bennett, David A Eichmann, Justin Guinney, Warren A Kibbe, Philip R O Payne, Emily R Pfaff, Peter N Robinson, Joel H Saltz, Heidi Spratt, Christine Suver, John Wilbanks, Adam B Wilcox, Andrew E Williams, Chunlei Wu, Clair Blacketer, Robert L Bradford, James J Cimino, Marshall Clark, Evan W Colmenares, Patricia A Francis, Davera Gabriel, Alexis Graves, Raju Hemadri, Stephanie S Hong, George Hripscak, Dazhi Jiao, Jeffrey G Klann, Kristin Kostka, Adam M Lee, Harold P Lehmann, Lora Lingrey, Robert T Miller, Michele Morris, Shawn N Murphy, Karthik Natarajan, Matvey B Palchuk, Usman Sheikh, Harold Solbrig, Shyam Visweswaran, Anita Walden, Kellie M Walters, Griffin M Weber, Xiaohan Tanner Zhang, Richard L Zhu, Benjamin Amor, Andrew T Girvin, Amin Manna, Nabeel Qureshi, Michael G Kurilla, Sam G Michael, Lili M Portilla, Joni L Rutter, Christopher P Austin, and Ken R, the N3C Consortium Gersing. "The National COVID Cohort Collaborative (N3C): Rationale, design, infrastructure, and deployment." *Journal of the American Medical Informatics Association*, (Oct 2020). ocaa196. https://doi.org/10.1093/jamia/ocaa196

[70] Cencheng Shen, and **Joshua T. Vogelstein**. "The exact equivalence of distance and kernel methods in hypothesis testing." *AStA Advances in Statistical Analysis*, (Sep 2020). https://doi.org/10.1007/s10182-020-00378-1

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FUNDING

EXTERNAL RESEARCH SUPPORT: CURRENT

2022-2027 **NIH, The Heart and the Mind: An Integrative Approach to Brain-Body Interactions (Data Core) [renewal of U19NS104653] U19NS104653**
PI: F. Engert · Role: Co-Investigator, Data Core Lead · Term: 01-Sep-2022 to 31-Aug-2027
Funding to lab, entire period: $3,085,025 (total, Data Core direct costs, per NIH RePORTER); current year: $205,569
Johns Hopkins University will be responsible for developing all algorithms and software in support of the Atlas project, as well as running the Data Core. This will include writing software to store, manage, and visualize the data, as well as algorithms for scalable analysis and support of modeling.

2020-2024 **NSF, Neural Net Learning for Graph NSF 2113099**, PI: C. Shen · Role: Co-Investigator · Term: 01-Sep-2021 to 31-Aug-2024
Funding to lab, entire period: $73,597 (total); current year:
Goal of this project is methodological development, theoretical investigation, and simulation and real data experimentation toward the end goal of principled understanding and advancement of the mathematics and science of graph neural network

2021-2025 **NIH, An Alignment Framework for Mapping Brain Dynamics and Substrates of Human Cognition Across Species 1RF1MH128696**
PI: T. Xu · Role: Co-Investigator · Term: 13-Sep-2021 to 12-Sep-2025
Funding to lab, entire period: $270,189 (total); current year:
We will continue collecting, organizing, and analyzing another cohort of the NKI-Rockland Sample.

2021-2027 **NIH, The NKI Rockland Sample II: An Open Resource of Multimodal Brain, Physiology & Behavior Data from a Community Lifespan Sample 2U19NS104653**
PI: M. Milham · Role: Co-Investigator · Term: 01-Jul-2022 to 30-Apr-2026
Funding to lab, entire period: $3,831,854 (total); current year:
The major goal is to establish multimodal MRI and electrophysiology lifespan sample to open and prospectively share with the larger scientific community.

2020-2027 **NSF, Collaborative Research: Transferable, Hierarchical, Expressive, Optimal, Robust, Interpretable Networks NSF 2031985**
PI: R. Vidal · Role: Co-Investigator · Term: 01-Sep-2020 to 31-Aug-2027
Funding to lab, entire period: $1,650,000 (direct); current year: $660,000 (direct)
The goal of this project is to develop a mathematical, statistical and computational frame- work that helps explain the success of current network arcitectures, understand its pitfalls, and guide the design of novel architectures with guaranteed confidence, robustness, inter- pretability, optimality, and transferability

2020 – **Microsoft, Federated Causal Inference for Multi-site Real-World Evidence & Clinical Trial Analysis Studies in Pandemic Preparedness**

PI: M. Powell · Role: Co-Investigator · Term: 01-Aug-2020 to current, Funding to lab, entire period: N/A; current year: N/A

This project will conduct federated retrospective analyses designed to assess the benefit of off-label drug use by pooling multiple disparate databases, to help prioritize and guide subsequent initiation and recruitment of randomized clinical trials. This will include evaluating the impact of the target drugs on patient outcomes from diseases similar to COVID-19, such as pneumonia or acute respiratory distress, generating artificial datasets using generative adversarial networks to asses performance of methods when 'ground truth' is known, applying the best methods to analyze the effect of the target drugs on the outcomes of COVID-19 patients across hospital systems, and using the results to evaluate the potential of these drugs and suggest guidelines for clinical trials.

2020-2023 **NIH, Graspy: A python package for rigorous statistical analysis of populations of attributed connectomes 1RF1MH123233**

PI: J. Vogelstein · Role: Principal Investigator · Term: 01-Jul-2020 to 30-Jun-2023

Funding to lab, entire period: $861,240 (direct) $1,416,279 (total); current year: $283,301 (direct) $471,082 (total)

The goal of this project is to establish a state-of-the-art toolbox for analysis of connectomes, spanning taxa, scale, and complexity. we will develop and extend implementations to enable neurobiologists to (1) estimate latent structure from attributed connectomes, (2) identify meaningful clusters among populations of connectomes, and (3) detect relationships be- tween connectomes and multivariate phenotypes

2019 – **Microsoft, Microsoft Research Award**, PI: J. Vogelstein · Role: Principal Investigator · Term: Unrestricted Gift

Funding to lab, entire period: $50,000 (total); current year: N/A

Research and development of neuroscience and connectomes around neuronal circuit and system modeling, application of time-series-of-graphs and dynamics to neuronal signaling analysis and connectomes, and in the abstractions of matter, math, machines that point toward complex systems composed of low-level components

EXTERNAL RESEARCH SUPPORT: COMPLETED

2020-2026 **NSF, NeuroNex2: Enabling Identification and Impact of Synaptic Weight in Functional Networks NSF 2014862**

PI: K. Harris · Role: Co-Investigator · Term: 15-Aug-2020 to 31-Jul-2026

Funding to lab, entire period: $609,294 (direct) $997,719 (total); current year: $121,587 (direct) $199,543 (total)

The goal is to develop the requisite technology to understand the impact of synaptic weight on functional networks

2020-2024 **NSF, AI Institute: Planning: From Biological Intelligence to Human Intelligence to Artificial General Intelligence (B2A) NSF 2020312**

PI: K. Kording · Role: Co-Investigator · Term: 01-Oct-2020 to 30-Sep-2024

Funding to lab, entire period: $500,000 (total); current year: N/A

The goal of this project is to plan an AI institution via several meetings and workshops

2020-2025 **NSF, CAREER: Foundational Statistical Theory and Methods for Analysis of Populations of Attributed Connectomes NSF 17-537**

PI: J. Vogelstein · Role: Principal Investigator · Term: 01-May-2020 to 30-Apr-2025

Funding to lab, entire period: $630,230 (total) $384,873 (direct); current year: N/A

The goal is to establish foundaitonal theory and methods for analyzing populations of attributed connectomes

2019-2025 **NIH, Brain Networks in Mouse Models of Aging NIH RO1AG066184**

PI: A. Badea · Role: Co-Investigator · Term: 15-Sep-2019 to 31-May-2025

Funding to lab, entire period: N/A; current year: $205,998

The goal of this grant is to generate connectomes and RNA-seq transcriptomes to characterize and differentiate APOE mice as a model of aging

2019-2022 **NIH, Accessible technologies for high-throughput, whole-brain reconstructions of molecularly characterized mammalian neurons NIH RFA-MH-19-148**

PI: M. Muller · Role: Co-Investigator · Term: 01-Sep-2019 to 31-Aug-2022

Funding to lab, entire period: $1,180,445 (total) $753,974 (direct); current year: $383,482 (total) $251,325 (direct)

The overall goal of the proposal is to develop technologies for the brain wide reconstruction of axonal arbors of molecularly defined neurons. The proposal aims at overcoming barriers in neuronal labeling, imaging and computation to achieve this goal, and to develop a technology platform that can be scaled to all neurons of the brain

2019-2025 **NIH, Reproducible imaging-based brain growth charts for psychiatry NIH R01MH120482**
PI: T. Satterthwaite · Role: Co-Investigator · Term: 22-Aug-2019 to 31-May-2025
Funding to lab, entire period: $362,861 (total) $231,276 (direct); current year: N/A
Aggregate, harmonize, and analyze existing large-scale pediatric neuroimaging datasets to identify normative and clinical brain growth curves

2018 – 2021 **NSF, SemiSynBio: Collaborative Research: YeastOns: Neural Networks Implemented in Communication Yeast Cells NSF 1807369**
PI: E. Schulman · Role: Co-Investigator · Term: 16-Jul-2018 to 30-Jun-2021
Funding to lab, entire period: $263,942 (total) $172,971 (direct); current year: $87,980 (total) $57,657 (direct)
Provide neuroscience and machine learning expertise to guide the design of the computa- tional learning capabilities of the system

2018 – 2020 **Schmidt Science Foundation, Connectome Coding at the Synaptic Scale Nascent Innovation Grant 128503**
PI: J. Vogelstein · Role: Principal Investigator · Term: 01-Jan-2018 to 31-Dec-2020
Funding to lab, entire period: $250,000 (total); current year: N/A
Study learning and plasticity at an unprecedented scale, revealing the dynamics of large populations of synapses comprising an entire local cortical circuit. No previously conducted experiment could answer the questions about the dynamics of large populations of synapses, which is crucial to understanding the learning process

2017 – 2020 **NSF, NeuroNex Innovation Award: Towards Automatic Analysis of Multi-Terabyte Cleared Brains NSF 1707298**
PI: J. Vogelstein · Role: Principal Investigator · Term: 01-Sep-2017 to 31-Aug-2020 (No Cost Extension)
Funding to lab, entire period: $959,999 (total) $588,758 (direct); current year: $320,000 (total) $196,252 (direct)
We propose to lower the barrier to connecting data to analyses and models by providing a coherent cloud computational ecosystem that minimizes current bottlenecks in the scientific process

2017 – 2022 **NIH, Sensorimotor processing, decision making, and internal states: towards a realistic multiscale circuit model of the larval zebrafish brain [original competitive segment of U19NS104653; renewed as "The Heart and the Mind"] NIH 1U19NS104653-01**
PI: F. Engert · Role: Co-Investigator · Term: 01-Sep-2017 to 31-Aug-2022
Funding to lab, entire period: $1,050,000 (total) $655,206 (direct) (JHU sub-award); current year: $210,000 (total) $131,041 (direct)
Generate a realistic multiscale circuit model of the larval zebrafish's brain – the multiscale virtual fish (MSVF). The model will span spatial ranges from the nanoscale at the synaptic level, to local microcircuits to inter-area connectivity - and its ultimate purpose is to explain and simulate the quantitative and qualitative nature of behavioral output across various timescales

2017 – 2020 **NIH, CRCNS US-German Res Prop: functional computational anatomy of the auditory cortex NIH 1R01DC016784-01**
PI: J. Ratnanather · Role: Co-Investigator · Term: 01-Jul-2017 to 30-Jun-2020
Funding to lab, entire period: $747,143 (total) $458,519 (direct); current year: N/A
Create a robust computational framework for analyzing the cortical ribbon in a specific region: the auditory cortex

2017 – 2020 **NSF, Multiscale Generalized Correlation: A Unified Distance-Based Correlation Measure for Dependence Discovery NSF 1921310**
PI: S. Cencheng · Role: Co-Investigator · Term: 01-May-2017 to 30-Apr-2020
Funding to lab, entire period: $200,000 (total) $124,189 (direct); current year: N/A
Establish a unified methodology framework for statistical testing in high-dimensional, noisy, big data, through theoretical advancements, comprehensive simulations, and real data experiments

2017 – 2019 **NSF, NeuroNex Technology Hub: Towards the International Brain Station for Accelerating and Democratizing Neuroscience Data Analysis and Modeling NSF 16-569**
PI: J. Vogelstein · Role: Principal Investigator · Term: 2017 to 2019
Funding to lab, entire period: $246,773; current year: N/A
We propose to lower the barrier to connecting data to analyses and models by providing a coherent cloud computational ecosystem that minimizes current bottlenecks in the scientific process

2017 – 2018 **The Kavli Foundation, The International Brain Station 90071826**
PI: J. Vogelstein · Role: Principal Investigator · Term: 2017 to 2018
Funding to lab, entire period: $50,000 (total) $50,000 (direct); current year: N/A
Take the first few steps towards building the international brain station

2017 – 2021 **DARPA, Lifelong Learning Forests FA8650-18-2-7834**

PI: J. Vogelstein · Role: Principal Investigator · Term: 01-Nov-2017 to 31-Oct-2021

Funding to lab, entire period: $1,839,308 (total) $1,123,474 (direct); current year: $199,179 (total) $121,667 (direct)

Lifelong Learning Forests (L2Fs) will learn continuously, selectively adapting to new environ- ments and circumstances utilizing top-down feedback to impact low-level processing, with provable statistical guarantees, while maintaining computational tractability at scale

2017 – 2021 **DARPA, Continual Learning Across Synapses, Circuits, and Brain Areas FA8650-18-2-7834**

PI: A. Tolias · Role: Co-Investigator · Term: 01-Nov-2017 to 30-Oct-2021

Funding to lab, entire period: $796,715 (total) $486,666 (direct); current year: $199,179 (total) $121,667 (direct)

Develop the pre-processing analysis pipeline for the imaging data collected in this project

2017 – 2018 **DARPA, The Brain Ark 90076467**, PI: J. Vogelstein · Role: Principal Investigator · Term: 2017 to 2018

Funding to lab, entire period: $92,376 (total) $56,499.08 (direct); current year: N/A

Characterize the statistical properties of the individual graphs, to identify circuit motifs, both that specialize in a species specific fashion, and that are preserved across species. As a test, will compare the connectomes of sea lions and coyotes

2017 – 2018 **NSF, Brain Comp Infra: EAGER: BrainLab CI: Collaborative, Community Experiments ACI-1649880**

PI: B. Miller · Role: Co-Investigator · Term: 2017 to 2018

Funding to lab, entire period: $294,599 (total) $180,736 (direct); current year: N/A

The BrainLab CI prototype system will deploy an experimental-management infrastruc- ture that allows users to construct community-wide experiments that implement data and metadata controls on the inclusion and exclusion of data

2016 – 2019 **NSF, A Scientific Planning Workshop for Coordinating Brain Research Around the Globe NSF 1637376**

PI: J. Vogelstein · Role: Principal Investigator · Term: 15-Apr-2016 to 31-Mar-2019

Funding to lab, entire period: $16,327 (total) $14,491 (direct); current year: N/A

This travel grant is for the expressed purposes of gathering researchers from around the globe to further discuss advancements in brain research during the second part of a two day conference

2016 – 2019 **NSF, A Scientific Planning Workshop for Coordinating Brain Research Around the Globe NIH RFA-MH-19-148**

PI: J. Vogelstein · Role: Principal Investigator · Term: 2016 to 2019

Funding to lab, entire period: $97,950 (total) $97,950 (direct); current year:

This travel grant is for the expressed purposes of gathering researchers from around the globe to discuss the new way to further brain research during part one of a two day conference

2016 – 2020 **DARPA, D3M: What Would Tukey Do? FA8750-17-2-0112**, PI: C. Priebe · Role: Co-Investigator · Term: 01-Oct-2016 to 30-Sep-2020

Funding to lab, entire period: $4,406,360 (total) $2,746,050 (direct); current year: N/A

Develop theory and methods for generating a discoverable archive of data modeling primi- tives and for automatically selecting model primitives and for composing selected primitives into complex modeling pipelines based on user-specified data and outcome(s) of interest

2015 – 2018 **DARPA, From RAGs to Riches: Utilizing Richly Attributed Graphs to Reason from N66001-15-C-40401**

PI: J. Vogelstein · Role: Principal Investigator · Term: 2015 to 2018

Funding to lab, entire period: $2,103,091.60 (total) $1,298,204 (direct); current year: N/A

Multiple, large, multifarious brain imaging datasets are rapidly becoming standards in neuroscience. Yet, we lack the tools to analyze individual datasets, much less populations thereof. Therefore, we will develop theory and methods to analyze and otherwise make such data available

2014 – 2016 **DARPA, Scalable Grain Graph Analyses Using Big-Memory, High-IPS Compute Architectures N66001-14-1-4028**

PI: R. Burns · Role: Co-Investigator · Term: 2014 to 2016

Funding to lab, entire period: $39,882 (total) $28,272 (direct); current year: N/A

Build software infrastructure to enable analytics on billion node, terabyte sized networks using commodity hardware

2014 – 2019 **NIH, Synaptomes of Mouse and Man NIH R01NS092474**, PI: S. Smith · Role: Co-Investigator · Term: 2014 to 2019

Funding to lab, entire period: $756,417 (total) $491,341 (direct); current year: N/A

The major goals of this project are to discover the synaptic diversity and complexity in mammalian brains, specifically comparing and contrasting humans with mice, the leading experimental animal

2012 – 2015 **National Institute of Biomedical Imaging and Bioengineering, CRCNS: Data Sharing: The EM open Connectome Project RO1EB16411**
PI: R. Burns · Role: Co-Investigator · Term: 2012 to 2015
Funding to lab, entire period: $70,823 (total) $46,517 (direct); current year: N/A
Develop cyberinfrastructure to support management, visualization, storage, and analysis of large-scale electron microscopy data

TALKS

INVITED TALKS

"Continual Learning Workshop (accepted, scheduled)." Columbia University, New York, NY, USA, 2026.

"Panelist, Careers in NeuroAI Professional Development Workshop (accepted, scheduled)." NIH BRAIN Initiative, Society for Neuroscience Annual Meeting, USA, 2026.

"Keynote Speaker: NeuroConnect 2025: Advancing Brain Network Research Workshop." NeuroConnect 2025: Advancing Brain Network Research Workshop, University of North Carolina, USA, 2025.

"It's about time: Learning for a Dynamic World." MATRIX AI Consortium for Human Well-Being, University of Texas San Antonio, USA, 2025.

"Learning for a Dynamic Future." Kavli NDI, Johns Hopkins University, USA, 2025.

"Learing when the future is partially predictable." University of Washington, Seattle, WA, USA, 2024.

"Learing when the future is partially predictable." Allen Institute for Brain Science, Seattle, WA, USA, 2024.

"Prospective Learning: Learning for a dynamic future." Cold Spring Harbor Laboratory, NY, USA, 2024.

"Moderator, SNF Global Center Third Expert Gathering: Consensus Building for Scalable Biometric Tools and Measures to Facilitate Global Mental Health Research." Child Mind Institute, NY, USA, 2024.

"Fixing Artificial Intelligence via Natural Intelligence (YouTube) ." Nemertes Next, Royal Sonesta Harbor Court, Baltimore, MD, USA, 2024.

"Learning the way that animals leaned to learn." Schizophrenia Center, Johns Hopkins University, Baltimore, MD, USA, 2024.

"Surprise! IID++ Out of Distribution & Prospective Learning." Simons Foundation, New York, NY, 2023.

"Hypothesis testing for connectome comparisons: a statistical analysis of bilateral symmetry in an insect brain connectome." Drexel University, Philadelphia, PA, 2022.

"Eliminating Accidental Deviations in Human Connectomics." JHU BME, Baltimore, MD, USA, 2021.

"Omnidirectional Lifelong Learning via Ensembling Representations." JHU BME, Baltimore, MD, USA, 2021.

"FIRM Guiding Principles for scientific software development and stewardship." JHU BME, Baltimore, MD, USA, 2021.

"Heritablity of Human Structural Connectomes." JHU BME, Baltimore, MD, USA, 2021.

"Towards a theory of out-of-distribution learning." JHU BME, Baltimore, MD, USA, 2021.

"Jovo++." JHU BME, Baltimore, MD, USA, 2021.

"Reality Transurfing: Chapter 1." JHU BME, Baltimore, MD, USA, 2021.

"Lifelong Learning: Theory and Practice." Darpa L2M PI Meeting, 2021.

"Lifelong Learning and Beyond." Darpa L2M PI Meeting, 2021.

"Lifelong Learning: Theory and Context." Darpa L2M PI Meeting, 2021.

"Lifelong Learning: Theory and Practice and Coresets." Darpa L2M PI Meeting, 2021.

"Lifelong Learning." North Carolina State University, Raleigh, NC, USA, 2020.

"Lifelong Learning." Morgan State University, Baltimore, MD, USA, 2020.

"Lifelong Learning: Moving Beyond Avoiding Catastrophic Forgetting." Johns Hopkins Mathematical Institute for Data Science, Baltimore, MD, USA, 2020.

"Open Access to the Brain: a Computer "Connectome" Links Brain Images in Fine Detail." JHM Boot Camp, Baltimore, MD, USA, 2019.

"Big Biomedical Data Science." Sol Goldman International Conference, Baltimore, MD, USA, 2019.

"Journey to Here." JHU BMES talks, Baltimore, MD, USA, 2019.

"NeuroData (Science)." Kavli, Baltimore, MD, USA, 2019.

"NeuroData Tools." NeuroData Hackashop, Baltimore, MD, USA, 2019.

"Biomedical Big Data and Data Science." JHU BME, Baltimore, MD, USA, 2019.

"Data Intensive Brain Science." Kavli Neuroscience Discovery Institute, Baltimore, MD, USA, 2018.

"Using Big Data Science to Understand What Goes On in our Heads." SOHOP Faculty Spotlight, Baltimore, MD, USA, 2018.

"Engineering the Future of Medicine: Data Intensive Biomedical Science." Johns Hopkins University Biomedical Engineering, Baltimore, MD, USA, 2018.

"Data Coordination and Data Resources for the BRAIN Initiative." 4th Annual BRAIN Initiative Investigators Meeting, Rockville, MD, USA, 2018.

"The International Brain Station (TIBS)." JHU BME and Tsinghua University, Baltimore, MD, USA, 2017.

"Opportunities and Challenges in Big Data Neuroscience." Society for Neuroscience, Washington D.C., USA, 2017.

"Using Big Data Science to Understand What Goes on in Our Heads." SOHOP Faculty Spotlight, Baltimore, MD, USA, 2017.

"NeuroStorm." Global Brain Workshop 2 JHU, Baltimore, MD, USA, 2017.

"Challenges and Opportunities in Big Data for Neuroscientists." Society for Neuroscience: DC Metro Area Chapter Keynote Address, Washington, DC, USA, 2017.

"The International Brain Station (TIBS)." United Nations Global Brain Workshop Meeting, Baltimore, MD, USA, 2016.

"Using Big Data Science to Understand What Goes on in Our Heads." SOHOP Faculty Spotlight, Baltimore, MD, USA, 2016.

"The International Brain Station (TIBS)." Kavli Foundation, Baltimore, MD, USA, 2016.

"NeuroData 2016." NeuroData Lab Retreat, 2016.

"Global Brain Workshop 2016." Global Brain Workshop NSF+JHU at Kavli, Baltimore, MD, USA, 2016.

"Global Brain Workshop 2016." Kavli Neuroscience Discovery Institute & Center for Imaging Science, Baltimore, MD, USA, 2016.

"Global Brain Workshop 2016." Kavli Neuroscience Discovery Institute & Center for Imaging Science @ JHU, Baltimore, MD, USA, 2016.

"Learning a Data-Driven Nosology:Progress, Challenges & Opportunities." Kavli Neuroscience Discovery Institute & Center for Imaging Science, Baltimore, MD, USA, 2016.

"NeuroData:Enabling Terascale Neuroscience." Kavli Neuroscience Discovery Institute & Center for Imaging Science, Baltimore, MD, USA, 2016.

"NeuroData:Enabling Terascale Neuroscience." JHU Kavli Neuroscience Discovery Institute, Baltimore, MD, USA, 2016.

"Special Symposium: Neuroscience in the 21st Century." Kavli, Baltimore, MD, USA, 2015.

"Open Connectome Project: Lowering the Barrier to Entry of Big Data Neuroscience." Institute for Computational Medicine at Johns Hopkins University, Baltimore, MD, USA, 2015.

"Open Source Platform for Heterogenous Brain Data." figshare, 2015.

"Big (Neuro) Statistics." Kavli Salon, Chicago, IL, USA, 2014.

"Open-Science Platform for Heterogeneous Brain Data: Opportunities and Challenges." Kavli, Baltimore, MD, USA, 2014.

"Big (Neuro) Statistics." Kavli Salon, Baltimore, MD, USA, 2014.

"Decision Theoretic Approach to Statistical Inference." Guest Lecture in Current Topics in Machine Learning, Johns Hopkins University, Baltimore, MD, USA, 2012.

"Once we get connectomes, what the %#* are we going to do with them?." Institute of Neuroinformatics, Boston, MA, USA, 2011.

"Inferring spike times given typical time-series fluorescence observations." Department of Applied Mathematics and Statistics, Johns Hopkins University, Baltimore, MD, USA, 2008.

OTHER TALKS

"Learning: the next frontier." National University of Public Service, Budapest, Hungary, 2024.

"Learning the way that animals leaned to learn." Department of Biomedical Engineering, McGill University, Canada, 2024.

"Probably Approximately Correct in the Future." Banff International Research Station for Mathematical Innovation and Discovery, Oaxaca, Mexico, 2024.

"Why do networks need negative weights?." , 2022.

"Simulating a Realization of a Stochastic Block Model." ABCD-ReproNim Program, 2022.

"Community Detection and Model Selection in SBMs." ABCD-ReproNim Program, 2022.

"Nonparametric MANOVA via Independence Testing." Global Young Scientists Summit, 2022.

"Maggot brain, mirror image? A statistical analysis of bilateral symmetry in an insect brain connectome." Neuromatch 4.0, 2021.

"graspologic: A python package for rigorous statistical analysis of populations of attributed connectomes." BRAIN Informatics Webinar, 2021.

"Nonparametric MANOVA via Independence Testing." BRAIN Initiative Meeting, 2021.

"Network data science for bilateral brains: Applications in the larval Drosophila connectome." NIH & DOE Brain Connectivity Workshop Series, 2021.

"OOD DARPA Presentation." DARPA, 2021.

"Lifelong Learning and Beyond." DARPA L2M, 2021.

"The role of the connectome in achieving artificial general intelligence." Yale School of Medicine, Whistler Scientific Workshop, Whistler, BC, Canada, 2020.

"Lifelong Learning via Ensembling General Representations." , 2020.

"A Theory and Practice of Lifelong Learnable Forest." Kavli Neural Systems Institute, Rockefeller University, New York City, NY, USA, 2020.

"Lifelong Learning." Columbia University, New York City, NY, USA, 2020.

"Ailey in an Hour: (A "Soup-to-Nuts" Pipeline for Analysis of Whole Cleared Brain Data)." NeuroNex, Cornell University, Ithaca, NY, USA, 2019.

"A Theory and Practice of the Lifelong Learnable." L2M, 2019.

"Data Science Core." Harvard University, Carmridge, MA, USA, 2019.

"Statistical Methods for Population of Connectomes." Organization of Human Brain Mapping, Rome, Italy, 2019.

"Forest Packing: Fast Parallel, Decision Forests." SIAM International Conference on Data Mining, Calgary, Alberta, Canada, 2019.

"Brain mapping tools for neuroscience research." NeuroNex, Cornell University, Ithaca, NY, USA, 2019.

"Big Data and the Life Sciences." Sloan Foundation, New York City, NY, USA, 2019.

"Statistical Foundations For Connectomics." Max Planck / HHMI Connectomics Meeting, Berlin, Germany, 2019.

"Connectal Coding." Dipy Workshop, Bloomington, Indiana, USA, 2019.

"Lifelong Learning Forests." L2M, 2019.

"Connectome Coding." Society for Neuroscience, San Diego, CA, USA, 2018.

"NeuroData: A Community-developed open-source computational ecosystem for big neuro data." NeuroNex, Cornell University, Ithaca, NY, USA, 2018.

"A Community-Developed Open-Source Computational Ecosystem for Big Neuro Data." Princeton University, Princeton, NJ, USA, 2018.

"Multiscale Graph Correlation: A Knowledge Representation System for Discovering Latent Geometric Structure." DARPA SIMPLEX PI Review Meeting, New York City, NY, USA, 2018.

"NeuroData: Embracing Open Source for Big Data Neuroscience." NSF NeuroNex Workshop on Super 3DEM, Austin, TX, USA, 2018.

"A High-Throughput Pipeline Identifies Robust Connectomes but Troublesome Variability." Organization of Human Brain Mapping, Suntec, Singapore, 2018.

"NeuroNex + Stanford." NeuroNex-Stanford, Stanford, CA, USA, 2018.

"Connectome Coding: what is it, how do we do it, and why do we care?." Data science in Neuroscience Symposium, Suntec, Singapore, 2018.

"Lifelong Learning Forests." Darpa L2M PI Meeting, Arlington, VA, USA, 2018.

"Discovering Relationships and their Geometry Across Disparate Data Modalities." Yale University, New Haven, CT, USA, 2018.

"Connectome Coding." Schmidt Sciences, 2017.

"Discovering Relationships and their Geometry Across Disparate Data Modalities." Stanford University, Stanford, CA, US, 2017.

"knor: a NUMA-Optimized In-Memory, Distributed and Semi-External-Memory k-means library." HPDC, Washington DC, USA, 2017.

"Science in the Cloud (SIC): A use-case in MRI Connectomics." Open Science Special Interest Group, Oxford University, Oxford, England, 2017.

"Network Dependence Testing via Diffusion Maps and Distance-Based Correlations." Joint Statistical Meetings, Baltimore, MD, USA, 2017.

"ROFLMAO: Robust Oblique Forests with Linear Matrix Operations." SIAM International Conference on Data Mining, Houston, TX, USA, 2017.

"NeuroData: Enabling Terascale Neuroscience for Everyone." 3rd Annual BRAIN Iniative Investigators Meeting, Bethesda, MD, USA, 2016.

"Multiscale Generalized Correlation." Joint Statistical Meeting, Chicago, IL, USA, 2016.

"NeuroData: Enabling Terascale Neuroscience for Everyone." Keystone Symposia: State of the Brain, Alpbach, Austria, 2016.

"Local Distance Correlation for Testing Independence." Temple University, Philadelphia, PA, USA, 2015.

"Law of Large Graphs." DARPA Graphs, Columbia University, New York City, NY, USA, 2015.

"big time (series data in neuroscience)." figshare, 2015.

"Research Computing Support for Neuroscience and Other Life Sciences." CASC, Aachen, Germany, 2015.

"From RAGs to Riches: Utilizing Richly Attributed Graphs to Reason from Heterogeneous Data." SIMPLEX Kickoff, New York City, NY, USA, 2015.

"From RAGs to Riches: Utilizing Richly Attributed Graphs to Reason from Heterogeneous Data: Part 1." DARPA SIMPLEX PI Meeting, New York City, NY, USA, 2015.

"From RAGs to Riches: Utilizing Richly Attributed Graphs to Reason from Heterogeneous Data: Part 2." DARPA SIMPLEX PI Meeting, New York City, NY, USA, 2015.

"Spike inference from calcium imaging using sequential Monte Carlo methods." AMSI Program on Sequential Monte Carlo, 2015.

"Opportunities and Challenges in Big Data Neuroscience." DoE, 2015.

"Top Challenges of Big Data Neuroscience." BRAIN Initiative Workshop, Bethesda, MD, USA, 2014.

"Big Statistics for Brain Sciences." Baylor College of Medicine, Department of Neuroscience, Houston, TX, USA, 2014.

"Beyond Little Neuroscience." Beyond Optogenetics workshop at Cosyne, Salt Lake City, UT, USA, 2013.

"Statistical Inference on Graphs." University of Michigan, Ann Arbor, Michigan, 2013.

"Statistical Inference on Graphs." Scientific Computing Institute, University of Utah, Salt Lake City, UT, USA, 2013.

"Open Problems in Neuropsychiatry." Data Seminar, Duke University, Durham, NC, USA, 2013.

"Statistical Models and Inference for big Brain-Graphs." NIPS Workshop on Acquiring and analyzing the activity of large neural ensembles, Lake Tahoe, NV, USA, 2013.

"BIG NEURO." Theory and Neurobiology, Duke University, Durham, NC, USA, 2012.

"Open Connectome Project." Academic Medical Center, Amsterdam, Netherlands, 2012.

"Are mental properties supervenient on brain properties." , 2011.

"What can Translational neuroimaging Research do for Clinical Practice." Child Mind Institute, New York City, NY, USA, 2011.

"Statistical Connectomics." Harvard University Connectomics Labs, Cambridge, MA, USA, 2011.

"Once we get connectomes, what the %#* are we going to do with them?." Krasnow Institute for Advanced Study at George Mason Univeristy, Fairfax, VA, USA, 2011.

"Consistent Connectome Classification." Math/Bio Seminar, Duke University, Durham, NC, USA, 2011.

"Connectome Classification: Statistical Graph Theoretic Methods for Analysis of MR-Connectome Data." Organization for Human Brain Mapping, Quebec City, Canada, 2011.

"Consistent Graph Classification." Guest Lecture in Deisseroth Lab, Stanford University, Stanford, CA, USA, 2011.

"Neurocognitive Graph Theory." National Security Agency, 2009.

"OOPSI: A Family of Optimal OPtical Spike Inference Algorithms for Inferring Neural Connectivity from Population Calcium Imagin g." Dissertation Defense, Johns Hopkins University, Baltimore, MD, USA, 2009.

"Sequential Monte Carlo in Neuroscience." SAMSI Program on Sequential Monte Carlo, Tracking Working Group, 2009.

"Towards Inference and Analaysis of Neural Circuits Inferred from Population Calcium Imaging." Guest Lecture in Schnitzer Lab, Stanford University, Stanford, CA, USA, 2009.

"Towards Inferring Neural Circuits from Calcium Imaging." Guest Lecture in Yuste Lab, Columbia University, New York City, NY, USA, 2009.

"Inferring Spike Trains Given Calcium-Sensitive Fluorescence Observations." Statistical Analysis of Neural Data, Pittsburgh, PA, USA, 2008.

"Inferring spike trains from Calcium Imaging." Redwood Center for Theoretical Neuroscience, University of California, Berkeley, CA, USA, 2008.

"Inferring spike trains from Calcium Imaging." Cambridge University, Gatsby Unit, and University College London, Cambridge, England, 2008.

"Model based optimal inference of spike times and calcium dynamics givern noisy and intermittent calcium-fluorescence observations." Neurotheory Center of Columbia University, New York City, NY, USA, 2007.

ABSTRACTS/POSTER PRESENTATIONS

ABSTRACTS / POSTERS

Ashwin De Silva, Rahul Ramesh, Rubing Yang, Siyu Yu, **Joshua T Vogelstein**, and Pratik Chaudhari. "Prospective Learning: Learning for a Dynamic Future." (Jul 2025).

Benjamin D Pedigo, Mike Powell, Eric W Bridgeford, Michael Winding, Carey E Priebe, and **Joshua T Vogelstein**. "Generative network modeling reveals a first quantitative definition of bilateral symmetry exhibited by a whole insect brain connectome." (Mar 2022). https://figshare.com/articles/poster/Generative_network_modeling_reveals_a_quantitative_definition_of_bilateral_symmetry_exhibited_by_a_whole_insect_brain_connectome/19610013

Jong M Shin, Leyla Isik, and **Joshua T Vogelstein**. "Measure of human-likelihood in tree-based ensemble model and artificial neural networks." (Mar 2022). https://figshare.com/articles/poster/2022_NAISys_conference_poster_presentation/20070515

Jayanta Dey, Will LeVine, Laknath A De Silva, Ali Geisa, and **Joshua T Vogelstein**. "Out-of-distribution Detection Using Kernel Density Polytopes." (Mar 2022). https://figshare.com/articles/poster/jayanta-NAISys2022_pdf/20070512

Javier J How, Gregor Schuhknecht, Misha B Ahrens, Florian Engert, and **Joshua T Vogelstein**. "Transfer learning in larval zebrafish (Danio rerio)." (Mar 2022). https://figshare.com/articles/poster/javier-NAISys2022_pdf/20070509

Laknath A De Silva, and **Joshua T Vogelstein**. "Kernel density networks." (Mar 2022). https://figshare.com/articles/poster/Kernel_Density_Networks/27245751?file=49842411

Haoyin Xu, and **Joshua T Vogelstein**. "Simplest streaming trees." (Mar 2022).

Joshua T. Vogelstein, Hayden Helm, Benjamin D. Pedigo, Ronak Mehta, Carey E. Priebe, and Chris White. "A Biological Implementation of Lifelong Learning in the Pursuit of Artificial General Intelligence." (Nov 2020).

Benjamin D Pedigo, Michael Winding, Turan Orujlu, Marta Zlatic, Albert Cardona, Carey E Priebe, and **Joshua T Vogelstein**. "A quantitative comparison of a complete connectome to artificial intelligence architectures." (Nov 2020).

Vivek Gopalakrishnan, Jaewon Chung, Eric Bridgeford, Jesus Arroyo, Benjamin D. Pedigo, Carey E. Priebe, and **Joshua T. Vogelstein**. "Statistical Methods for Multiscale Comparative Connectomics." *Imaging Neuroscience*, (Nov 2020). https://direct.mit.edu/imag/article/doi/10.1162/IMAG.a.2/130633/Multiscale-comparative-connectomics

Benjamin D. Pedigo, Michael Winding, Ali Saad-Eldin, Tingshan Liu, Albert Cardona, Marta Zlatic, Carey E. Priebe, and **Joshua T. Vogelstein**. "Statistical tools for nanoscale connectomics: clustering neurons in Drosophila larva brain and other applications." (Nov 2020).

Jaewon Chung, Jayanta Dey, Gregory Kiar, Carey E. Priebe, and **Joshua T. Vogelstein**. "Human Structural Connectomes are Heritable." (Nov 2020).

Ali Saad-Eldin, Benjamin D. Pedigo, Youngser Park, Carey E. Priebe, and **Joshua T. Vogelstein**. "NeuroGraphMatch." (Nov 2020).

Eric W. Bridgeford, Michael Powell, Anton Alyakin, Brian Caffo, and **Joshua T. Vogelstein**. "Batch Effects are Causal Effects: Applications in Human Functional Connectomes." (Nov 2020).

Ronan Perry, Jelle Zorn, Sebastien Czajko, Daniel S. Margulies, and **Joshua T. Vogelstein**. "Permutation-corrected independence testing for high-dimensional fMRI data." (Nov 2020).

Jin Hecheng, Julian S.B. Ramirez, **Joshua T. Vogelstein**, Michael P. Milham, and Ting Xu. "Assessing functional connectivity beyond Pearson's correlation." (Sep 2020).

X Li, J.W. Cho, Michael P. Milham, and Ting Xu. "Improving brain-behavior prediction using individual-specific components from connectivity-based shared response model." (Sep 2020).

J.W. Cho, A. Korchmaros, **Joshua T. Vogelstein**, Michael P. Milham, and Ting Xu. "Developing a gradient flow framework to guide the optimization of reliability for the study of individual differences." (Sep 2020).

J.W. Cho, A. Korchmaros, **Joshua T. Vogelstein**, Michael P. Milham, and Ting Xu. "Impact of Concatenating fMRI Data on reliability for Functional Connectomics." (Sep 2020).

Eric Bridgeford, and **Joshua T. Vogelstein**. "Optimal Experimental Design for Big Data: Applications in Brain Imaging." (Jun 2020).

Ronan Perry, and **Joshua T. Vogelstein**. "Identifying Differences Between Expert and Novice Meditator Brain Scans via Multiview Embedding." (Jun 2020).

J.W. Cho, A. Korchmaros, **Joshua T. Vogelstein**, Michael P. Milham, and Ting Xu. "Impact of Concatenating fMRI Data on reliability for Functional Connectomics." (Jun 2020).

J.W. Cho, A. Korchmaros, **Joshua T. Vogelstein**, Michael P. Milham, and Ting Xu. "Developing a gradient flow framework to guide the optimization of reliability for the study of individual differences." (Jun 2020).

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EDUCATIONAL ACTIVITIES

TEACHING EXPERIENCE - ONGOING COURSES

Spring '25 **Co-Instructor, JHU**, *EN.580.438/638*, *Biomedical Data Design II*, with Kwame Sanwu Kutten

Fall '24 **Course Director, JHU**, *EN.580.465*, *Natural Intelligence*

Fall '24 **Co-Instructor, JHU**, *EN.580.437/697*, *Biomedical Data Design*, with Kwame Sanwu Kutten

Spring '24 **Course Director, JHU**, *EN.580.402*, *Commitments of Conscience Leadership*

Fall '23 **Course Director, JHU**, *EN.580.419*, *Philosophy of Life: A Data Science Perspective*

Fall '22 **Co-Instructor, JHU**, *EN.580.564*, *Advanced Focus Area Research: Biomedical Data Science*, with Eileen B. Haase

Spring '22 **Course Director, JHU**, *EN.580.218*, *Learning Salon*

Spring '22 **Course Director, JHU**, *EN.580.238/438/638*, *NeuroData Design II*

Fall '21 **Co-Instructor, JHU**, *EN.580.564*, *Advanced Focus Area Research: Biomedical Data Science*, with Eileen B. Haase

Spring '21 **Course Director, JHU**, *EN.580.419*, *Philosophy of Life - A Data Science Perspective*, enrollment 6

Spring '22 **Course Director, JHU**, *EN.580.438/638*, *NeuroData Design II*, enrollment 20

Fall '21 **Course Director, JHU**, *EN.580.437/697*, *NeuroData Design I*, enrollment 27

Spring '21 **Course Director, JHU**, *EN.580.438/638*, *NeuroData Design II*, enrollment 30

Fall '20 **Course Director, JHU**, *EN.580.237/437/697*, *NeuroData Design I*, enrollment 38

Spring '20 **Course Director, JHU**, *EN.580.438/638*, *NeuroData Design II*, enrollment 32

Fall '19 **Course Director, JHU**, *EN.580.237/437/637*, *NeuroData Design I*, enrollment 46

Spring '19 **Course Director, JHU**, *EN.580.438/638*, *NeuroData Design II*, enrollment 18

Fall '18 **Course Director, JHU**, *EN.580.237/437/637*, *NeuroData Design I*, enrollment 22

Spring '17 **Course Director, JHU**, *EN.580.238/438/638*, *NeuroData Design II*, enrollment 14

Winter '17 **Course Director, JHU**, *EN.580.574*, *BME Research Intersession*, enrollment 6

Fall '17 **Course Director, JHU**, *EN.580.247/437/637*, *NeuroData Design I*, enrollment 15

Spring '16 **Course Director, JHU**, *EN.580.468*, *The Art of Data Science*, enrollment 24

Fall '16 **Course Director, JHU**, *EN.580.437*, *NeuroData Design I*, enrollment 16

Spring '15 **Course Director, JHU**, *EN.580.694*, *Statistical Connectomics*, enrollment 26

TEACHING EXPERIENCE - ONE-TIME

Spring '19 **Guest Lecturer, JHU**, *EN.580.422*, *Systems Bioengineering II*, 2 Lectures

Spring '19 **Guest Lecturer, JHU**, *AS.080.321*, *Computational Neuroscience*, 2 Lectures

Spring '18 **Guest Lecturer, JHU**, *EN.580.422*, *Systems Bioengineering II*, 2 Lectures

Spring '18 **Guest Lecturer, JHU**, *AS.080.321*, *Computational Neuroscience*, 2 Lectures

Spring '17 **Guest Lecturer, JHU**, *EN.580.422*, *Systems Bioengineering II*, 2 Lectures

Spring '16 **Guest Lecturer, JHU**, *EN.580.422*, *Systems Bioengineering II*, 2 Lectures

Winter '16 **Guest Lecturer, JHU**, *EN.600.221*, *Introduction to Connectomics*, 1 Lecture

Fall '16 **Guest Lecturer, JHU**, *EN.580.111*, *BME Modeling and Design*, 1 Lecture

Fall '15 **Course Co-Director, JHU**, *Introduction to Computational Medicine*

Educational Workshops

Summer '19 **DiPy Workshop**, *Bloomington, Indiana*, 1 day lecture on statistical connectomics

Fall '18 **Society for Neuroscience Annual Meeting**, *Educational Workshop*, *San Diego, CA*, 1 day lecture on statistical connectomics

Fall '17 **Society for Neuroscience Annual Meeting**, *Educational Workshop*, *San Diego, CA*, 1 day lecture on statistical connectomics

Summer '16 **CRCNS Course on Mining and Modeling of Neuroscience Data**, *Redwood Center for Theoretical Neuroscience University of California, Berkeley*, 2 day lecture on statistical connectomics

MENTORSHIP

RESEARCH TRACK FACULTY MENTORSHIP

07/19 – 08/20 **Ronak Mehta**, *Research Assistant · MSE · BME, JHU*
Finalizing three manuscripts on (1) uncertainty forests, (2) time-series dependence quantification, and (3) lifelong learning forests

03/19 – 05/20 **Anton Alyakin**, *Assistant Research Engineer · BSE · BME, JHU*, Worked on various problems in statistical graph inference

08/16 – 08/18 **Eric Perlman**, *Assistant Research Faculty · Ph.D. · BME, JHU*
Lead Scientist in developing storage, transfer, and visualization solutions for large data in our cloud infrastructure

03/16 – 06/20 **Jesse Patsolic**, *Assistant Research Faculty · MA · BME, JHU*
Lead developer converting our extensions to decision forests to be merged into sklearn

02/19 – 12/19 **Hayden Helm**, *Assistant Research Faculty · MSE · BME, JHU*
Lead research efforts developing theory and methods for lifelong learning

STAFF RESEARCH SCIENTISTS

10/23 – 2024 **Itsuki Ogihara**, *Research Assistant · BME · MS, JHU*

09/20 – 04/23 **Jong Shin**, *Software Engineer · MSE · BME, JHU*
Currently investigating the effect of inductive bias innately coinciding with various machine learning models

06/19 – 08/20 **Devin Crowley**, *Research Assistant · BS · BME, JHU*, Lead developer of our scalable Python implementation of LDDMM

06/18 – 12/19 **Benjamin Falk**, *Research Engineer · Ph.D. · BME, JHU*
Lead software engineer, oversees all development projects, solely responsible for all cloud infrastructure

03/20 – 08/22 **Ali Geisa**, *Research Assistant · MS · BME, JHU*, Researching progressive and lifelong learning theory

POSTDOCTORAL FELLOWS

08/14 – 05/22 **Tyler Tomita**, *Postdoctoral Fellow · MSE · BME, JHU*
Developed Sparse Projection Oblique Randomer Forest in his dissertation, currently the best performing machine learning algorithm on a standard suite of over 100 benchmark problems. Currenly a postdoc with Assistant Prof. Chris Honey of Psychology and Brain Sciences

11/20 – 11/25 **Javier Josue How**, *Postdoctoral Fellow · Ph.D. · Neurosciences, UCSD*
Javier studies how larval zebrafish learn how to perform a task under one situation, and use this knowledge to learn another task more quickly. He hopes to use this understanding of biological transfer learning to improve machine learning, which tends to be unable to complete this feat.

08/18 – 08/20 **Jesús Arroyo**, *Post-doctoral Fellow · Ph.D. · CIS, JHU*, Worked on graph matching and joint graph embedding

09/16 – 08/18 **Cencheng Shen**, *Post-doctoral Fellow · Ph.D. · CIS, JHU*
Developed Multiscale Graph Correlation, which is currently the premiere hypothesis testing framework, and about to be integrated into SciPy, by far the world's leading scientific computing package. Currently an Assistent Professor in Department of Statistics at University of Delaware, and still an actice collaborator and grantee

06/16 – 07/17 **Guilherme Franca**, *Post-doctoral Fellow · Ph.D. · CIS, JHU*
Worked on non-parametric clustering, with an article about to be accepted in PAMI, the leading machine learning journal. Currently a postdoc for Rene Vidal

07/19 – 08/21 **Celine Drieu**, *Post-doctoral Fellow · Ph.D. · Kavli NDI, JHU*
Co-Advised by Assitant Prof. Kuchibhotla, Department of Psychological and Brain Sciences. Working on understanding learning and memory using two-photon calcium imaging

07/18 – 07/20 **Audrey Branch**, *Post-doctoral Fellow · Ph.D. · Kavli NDI, JHU*
Co-Advised by Prof Michela Gallagher, extending brain clearing experimental technology from mice to rats. Currently with a manuscript on biorxiv

07/19 – 08/21 **Austin Grave**, *Post-doctoral Fellow · Ph.D. · Kavli NDI, JHU*
Co-Advised by Prof. Richard Huganir, Department of Neuroscience. Working on understanding whole brain synaptic plasticity using genetic engineering and light microscopy imaging

09/22 – 12/22 **Adam Li**, *Postdoctoral Fellow · Ph.D. · BME, JHU*

05/16 – 06/17 **Leo Duan**, *Post-doctoral Fellow · Ph.D. · CIS, JHU*
Went on to do a second postdoc with Leo Dunson (who I did my second postdoc with). Currently an Assistant Professor at University of Florida

PH.D. STUDENTS

01/21 – **Haoyin Xu**, *PhD Candidate · MSE · BME, JHU*
A Research Assistant who was also a Master's student in the NeuroData lab, maintainer of proglearn, working on streaming trees and forests

05/19 –- 07/24 **Jaewon Chung**, *PhD Student · MSE · BME, JHU*
Data science for macroscale connectomes. Co-creator and maintainer of `graspologic`, a Python package for network statistics.

08/19 – 12/23 **Eric Bridgeford**, *PhD Student · BSE · Department of Biostatistics, JHU*
Dissertation will focus on statistics of human connectomes and mitigating batch effects. Already first author on several manuscripts under review, and spearheads collaboration with Prof Brian Caffo at Biostatistics

08/18 – 05/23 **Ben Pedigo**, *PhD Candidate · BS · BME, JHU*
Data science for nanoscale connectomes. Co-creator and maintainer of `graspologic`, a Python package for network statistics.

07/19 – 06/24 **Jayanta Dey**, *PhD Student · MSE · BME, JHU*
Currently working on lifelong learning that aims at training a machine learning model on multiple tasks and transferring knowledge among tasks

05/20 – **Tingshan Liu**, *PhD Student · B.A. · Math Neuro, Smith College*
Implementing and applying clustering algorithms to the connectomes of inset nervous systems.

08/21 – **Ashwin De Silva**, *PhD Student · BS · BME, University of Moratuwa*, Statistical Machine Learning

02/22 – 05/25 **Alice Qingyang Wang**, *PhD candidate · Bsc · Neuroscience, JHU*

06/19 – 12/24 **Sambit Panda**, *PhD Student · MSE · BME, JHU*
A Ph.D. student who was also a Master's student in the NeuroData lab. Currently, the maintainer of the `hyppo` package, and works on creating more powerful and efficient multivariate hypothesis tests.

08/20 – 08/22 **Kaleab A. Kinfu**, *PhD Student · MSE · BME, JHU*
Kaleab studied double descent phenomena in decision forests and deep learning methods and developed 'Partition and Decode' – a framework that formalizes an implicit internal representation of several modern machine learning methods. He is currently a Ph.D. student in CS at JHU.

01/19 – 01/24 **Thomas Athey**, *PhD Candidate · BS · BME, JHU*
Tommy analyzes terabyte-scale full brain images from light microscopy with a focus on neuromorphology. His expertise is in statistics and computer vision.

05/23 – **Yuxin Bai**, *PhD Student · MSE · BME, JHU*

2024 – **Itsuki Ogihara**, *PhD Student · MS · BME, JHU*

7/2024 - **Cecelia Shuai**, *PhD student · Bachelor · Neuroscience, Johns Hopkins*
I'm interested in lifelong learning and am now working on prospective learning in the simulated ecological environment. I also model cool animal learning data collected in my joint-mentor's lab

09/24 - **Rodrigo Guerra**, *PhD Student · MS · BME, The City College of New York*, Psychedelics - all

8/23 - 12/23 **Skyler Thomas**, *PhD Student (Rotation) · BS, BSA · BME, JHU*
Skyler is a rotation student in the lab. He is interested in applied mathematics and machine learning. He is currently working on prospective learning theory. He is currently a Ph.D. student in BME at JHU.

01/22 – 01/23 **Noga Mudrik**, *PhD Student (Rotation) · Ph.D. · BME, JHU*

05/22 – 05/23 **Jeremy Welland**, *PhD Student (Rotation) · Ph.D. · BME, JHU*

08/16 – 12/21 **Vikram Chandrashekhar**, *PhD advisee · BSE · BME, JHU*
Dissertation has focused on extending LDDMM to whole cleared brain datasets, spearheads collaboration with Prof. Karl Deisseroth's lab at Stanford, one of the world's leading neuroscientists

08/18 – 06/2022 **Meghana Madyastha**, *PhD Co-advisee · BSE · CS, JHU*
Dissertation will focus on computational aspects of accelerationg learning and inference using decision forests

08/19 – 04/22 **Mike Powell**, *PhD Candidate · MSE · BME, Johns Hopkins University*
Mike has studied drug-repurposing options for potential COVID-19 treatments, proposed methodological improvements and best practices for drug-repurposing studies, and developed a taxonomy for describing and quantifying feature importance in machine learning models.

VISITING DOCTORAL STUDENT

03/19 – 09/19 **Derek Pisner**, *PhD advisee · MSE · JHU/UT, Austin*
Worked on the ndmg pipeline, developing direct streamline normalization for structural connectome generation

M.S. STUDENTS

6/23 – 8/24 **Ziyan Li**, *MS student · MSE · BME, JHU*

06/19 – 12/20 **Bijan Varjavand**, *MS advisee · BSE · BME, JHU*, Submitted manuscript to PAMI on advancing statistics on populaitons of networks

06/18 – 06/19 **Drishti Mannan**, *MS advisee · BSE · BME, JHU*
Preparing manuscript introducing novel specification for large attributed networks

10/18 – 04/22 **Alex Loftus**, *MS advisee · BSE · BME, JHU*, graph stats book, 'graspologic' package, ndmg development

06/19 – 05/21 **Vivek Gopalakrishnan**, *Combined BS/MSE Student · MSE · BME, JHU*
Vivek developed multiscale hypothesis tests for multi-subject connectomics datasets, and is currently a PhD student in Medical Engineering and Medical Physics at the Harvard-MIT Program in Health Sciences and Technology.

01/19 – 06/21 **Ronan Perry**, *MSE/BS Student · MSE · BME, JHU*
Ronan studied random forest methods for structured data and improved prediction calibration, developed nonparametric hypothesis testing tools, and explored novel complexity measures of neural networks. He is currently a Fulbright Germany scholar with Bernhard Scholkopf.

01/20 – 08/22 **Shreya Singh**, *Graduate Researcher · BS · BME, JHU*, brainlit' package, aws and azure data management

05/20 – 12/21 **Ali Saad-Eldin**, *MS advisee · BSE · BME, JHU*
Working on implementing and improving cominatorial optimization algorithms, specifically the Quadratic Assignment Problem

02/20 – 12/20 **Will LeVine**, *MS advisee · BME, JHU*
Exploring different sub-algorithms within progressive learning to alleviate harmful effects that resultfrom training on unhelpful data

08/14 – 06/17 **Greg Kiar**, *MSE advisee · BSE · BME, JHU*
Developer of m2g, the only existing "soup to nuts" pipeline for both functional and diffusion pipelines, co-first author of manuscript under review. Currently a PhD student at McGill University

07/19 – 04/22 **Ross Lawrence**, *MS advisee · BSE · BME, JHU*, Lead m2g developer, maintainer of neuroparc, MRI connectome repositories

09/24 – 08/25 **Siyu (Rice) Yu**, *MS student · MSE · BME, JHU*

11/24 - 08/25 **Z. Qin**, *MS student · MS · BME, JHU*

4/25 – **Beiya (Alexa) Xu**, *MS student · MSE · BME, JHU*

05/25 – **Yilin (Clark) Xu**, *MS Student · BS · BME, JHU*, Dedicated on Treeple acceleration

05/25 - **Yue (Zoe) Wan**, *MS student · BS · AMS, JHU*

UNDERGRADUATE STUDENTS

6/23 – 12/23 **Anvii Mishra**, *Undergraduate · BS · BME, JHU*

06/22 – 02/23 **Sejal Srivastava**, *Undergraduate · BS · BME, JHU*

09/21 – 08/24 **Kareef Ullah**, *Undergraduate Researcher · BME, JHU*, Assisted with fixing issues in graspologic and hyppo

10/22 – 05/25 **Hope Ugwuoke**, *Undergraduate · BS · BME, JHU*

06/22 – 12/22 **Audrey Herskovits**, *Undergraduate (Visiting) · BS · BME, JHU*

06/19 – 12/19 **Richard Guo**, *Undergraduate Researcher · BME, JHU*
Developed uncertainty forests, an approach for estimated posterior class probabilities, conditional entropy, and mutual information for high-dimensional data common in brain science applications

06/21 – 08/21 **Dominique Allen**, *Undergraduate Researcher · BME, JHU*, Assisted Thomas Athey in his work with statistics and computer vision

08/20 – 05/21 **Alisha Kodibagkar**, *Undergraduate Researcher · BME, JHU*, Assisting in the integration of brainlit packages with Azure services

05/15 – 05/16 **Jordan Matelsky**, *BSE · BME, JHU*
Currently a data scientist at APL, developed a number of simple WebApps in support of big data management

05/20 – 06/2022 **Diane Lee**, *Undergraduate Researcher · BME, JHU*, Assisting in the maintenance of graspologic

06/15 – 12/15 **Ron Boger**, *Undergraduate Researcher · BSE · BME, JHU*
Worked at a computational medicine start-up in Silicon Valley, worked on high-dimensional low-sample size theory

06/15 – 08/16 **Albert Lee**, *Undergraduate · BSE · BME, JHU*, Developed big data visualization tools

02/15 – 05/16 **Ivan Kuznetsov**, *BSE · BME, JHU*
Currently an MD, PhD Candidate at the UPenn, winner of Soros Fellowship, worked on analysis of data from Dr. Daniel Amen, developed matrix exploratory data analysis package.

HIGHSCHOOL STUDENT

07/22 – 08/22 **Meikah Robinson**, *High School Intern · BME, JHU*

Summer '19 **Shiyu Sun**, *Summer Intern · BME, JHU*, Applied to BME PhD Program in Fall 2020

Summer '19 **Sander Shulhoff**, *Summer Intern · BME, JHU*

Summer '19 **Kiki Zhang**, *Summer Intern · BME, JHU*

05/21 – 08/21 **MyCo Le**, *High School Intern · BME, JHU*

Summer '18 **Papa Kobina Van Dyck**, *Summer Intern · BME, JHU*, Applied to PhD Program in Fall 2019

THESIS COMMITTEE SERVICE (BME UNLESS NOTED OTHERWISE)

2026 **Ohata, Merrick**, *Applied Mathematics and Statistics*, *JHU Ph.D. Student*

2026 **Beatty, Kyle**, *Applied Mathematics and Statistics*, *JHU Ph.D. Student*

2026 **Chen, Tianyi**, *Applied Mathematics and Statistics*, *JHU Ph.D. Student*, GBO passed 2026

2025 **Chen, Qingyu**, *Biomedical Engineering*, *JHU Ph.D. Student*

2024 **Madhyastha, Meghana**, *Computer Science*, *JHU Ph.D. Student*, Graduated 2024

2024 – **Yang, Rubing**, *Applied Mathematics and Computational Science*, *University of Pennsylvania Ph.D. Student*
Defense scheduled Sept.\ 2026

2024 **Wang, Tim**, *Applied Mathematics and Statistics*, *JHU Ph.D. Student*, GBO 2024

2024 **Xie, Jiancheng**, *Biomedical Engineering*, *JHU M.S. Student*, Graduated 2024

2023 **Ramnauth, Anthony**, *Neuroscience*, *JHU Ph.D. Student*, Graduated 2023

2023 **Shapiro, Benjamin**, *Computer Science*, *JHU Ph.D. Student*, Graduated 2023

2019 **Browne, James**, *Computer Science*, *JHU Ph.D. Student*, Graduated 2019

2019 **Mhembere, Disa**, *Computer Science*, *JHU Ph.D. Student*, Graduated 2019

2018 **Kutten, Kwame**, *JHU Ph.D. Student*, *Graduated 2018*

2018 **Wang, Shangsi**, *Applied Mathematics and Statistics*, *JHU Ph.D. Student*, Graduated 2018

2018 **Tang, Runze**, *Applied Mathematics and Statistics*, *JHU Ph.D. Student*, Graduated 2018

2018 **Lee, Youjin**, *Biostatistics*, *JHU Ph.D. Student*, Graduated 2018

2017 **Zheng, D**, *Computer Science*, *JHU Ph.D. Student*, Graduated 2017

2017 **Binkiewicz, Norbert**, *Statistics*, *University of Wisconsin Ph.D. Student*, Graduated 2017

2016 **Gray-Roncal, Will**, *Computer Science*, *JHU Ph.D. Student*, Graduated 2016

SERVICE

GRANT REVIEW SERVICE

2025 **NIH Review Panel**, *TMM Study Section*

2015 **NSF Review Panel**, *Review for NSF BIG DATA Program*

UNIVERSITY SERVICE

Winter '19 **Track Organizer**, *AI in Healthcare: From Bench to Bedside*, *Organizer for Breakout Topic Sessions on artificial intelligence*

08/15 – 07/18 **Co-Developer**, *Computational Medicine Minor*, *JHU, Baltimore, MD, USA*

2015 – 2017 **Co-Founder and Faculty Advisor**, *MedHacks*
Medhacks is one of the first and largest hackathons dedicated specifically to hacking on medical advances, started entirely by BME undergrads at JHU

08/14 – 08/18 **Director of Undergraduate Studies**, *Institute for Computational Medicine*, *JHU, Baltimore, MD, USA*

DEPARTMENT SERVICE

2024 **Member**, *Promotion Committee*, *BME*, Jean Fan, promotion to Associate Professor with tenure

2019 **Member**, *Search Committee*, *BME*, Neuroengineering, 2019

2019 **Member**, *Search Committee*, *BME*, Data Science, 2019

2018 **Member**, *Search Committee*, *BME*, Neuroengineering, 2018

SERVICE IN SCIENTIFIC COMMUNITY

2019 – 2022 **Mentor**, *Black in AI*

2017 – **Scientific Advisory Board**, *NSF NeuroNex*
Enhanced resolution for 3DEM analysis of synapses across brain regions and taxa. Provided scientific, computational, and statistical guidance to a flagship NSF funded BRAIN Initiative program

2017 – **Chair of Committee of Data Cores**, *U19 Data Cores*
The U19 program is NIH's flagship BRAIN Initative program, with five original awardees, each with a dedicated Data Core and designated PI. I was elected the chair of the committee of Data Core PIs

2017 **Consultant for Nature Publishing Group**
The journal Nature, flagship journal of Nature Publishing Group, decided to create a "Code and Software Submission Checklist". They consulted me on their first draft, and I helped re-write it. An image of the final checklist is available here

2011 – **Open Connectome Project**
The co-founder of the "Open Connectome Project" (OCP), for several years, I was the only neuroscientist that could easily store, manage, and analyze very big datasets, spanning first tens of terabytes, and then hundreds. For that reason, I was an essential co-author on a number of big data papers. Specifically, though I sometimes contributed relatively little to the scientific ideas, I often was required to complete, visualize, and/or share the data. Perhaps more importantly, both funding agencies and journals began mandating that these large datasets be publicly shared, and OCP was literally the only option. This is despite often not having funding, nor being a co-author, on the manuscripts

2010 – **AWS Open Neuro Data Registry**
Our lab co-founded the Registry of Open Data on Amazon Web Serivces (AWS). The implication of this is that now, pending a few minor considerations, any neuroscientist that collects large image data can deposit it online *for free*. This means that neither they nor we must request funding to store the data. Our lab maintains this repository, but only by virtue of ensuring instructions for uploading, visualizing, and downloading are up to date, and acting as a gatekeeper to ensure only appropriate data are deposited there

JOURNAL SERVICE

Editorial Board

2019 – **Associate Editor**, *Journal of the American Statistical Association*

2018 – **Editor**, *Neurons, Behavior, Data analysis, and Theory*

2016 **Guest Associate Editor**, *PLoS Computational Biology*

Conference and Journal Reviewer

Proceedings of the National Academy of Sciences (PNAS)

International Conference on Artificial Intelligence and Statistics (AISTATS)

ACM SIGKDD Conference on Knowledge Discovery and Data Mining

2026 **Area Chair**, *Neural Information Processing Systems (NeurIPS)*

Neural Information Processing Systems (NeurIPS)

2026 **Program Committee**, *Brain Informatics*

Science Advances

Scientific Reports

Journal of Brain and Spine

BMC Biology

Journal of Computational Neuroscience
Molecular Brain
Annals of Clinical and Translational Neurology
Communications in Statistics – Simulation and Computation
Quantitative Imaging in Medicine and Surgery (QIMS)
PeerJ Computer Science
Briefings in Bioinformatics
Cell Biomaterials
International Conference on Learning Representations (ICLR)
International Conference on Machine Learning (ICML)
Annals of Applied Statistics (AOAS)
Bioinformatics
IEEE Global Conference on Signal and Information Processing (GlobalSIP)

CONFERENCES AND HACKATHON ORGANIZER

Summer '20 **Co-Chair**, *SciPy mini-symposium: Biology and Bioinformatics*

Winter '19 **Organizer**, *Decision Forest Hackathon*

Summer '19 **Organizer**, *NeuroData Workshop*, *https://neurodata.devpost.com*
Hackashop to train brain scientists in machine learning for big data (50) participants from around the country

March '19 **Organizer**, *Neuro Reproducibility Hackashop*, *https://brainx3.io/*
Hackashop to train brain scientists in best practices in reproducible science, co-organized with two startups: Vathes, LLC and Gigantum (50 participants)

Spring '18 **Organizer**, *NeuroData Hackathon*

Fall '17 **Organizer**, *NeuroData Mini-Hackathon*

Summer '17 **Organizer**, *NeuroStorm*, *https://brainx2.io*
Workshop to bring together thought leaders from academia, national labs, industry, and non-profits around the world to take next steps towards accelerating brain science discovery in the cloud (50 participants and 5 observers from funding institutions)

2016 **Organizer**, *Global Brain Workshop*, *http://brainx.io*
First ever international Brain Initiative workshop, bringing together leaders from around the world, covered by Nature and Science (75 participants)

2016 **Co-Organizer**, *Brains and Bits: Neuroscience Meets Machine Learning, NIPS Workshop*
http://www.stat.ucla.edu/ akfletcher/brainsbits_overview.html

Winter '15 **Organizer**, *Hack@NeuroData*, *http://hack.neurodata.io/*

2015 **Co-Organizer**, *BigNeuro2015: Making Sense of Big Neural Data, NIPS Workshop*, *http://neurodata.io/bigneuro2015*

2012 **Co-Organizer**, *Scaling up EM Connectomics Conference*
The world's first connectomics workshop, now run annually alternating between Janelia Research and Max Plank locations (80 participants)

AWARDS AND RECOGNITION

INDIVIDUAL

2002 **Dean's List**, *Washington University*

SHARED (12)

2025 **OpenCog Foundation Best Student Paper Award, AGI-25**, *Bai, Shuai, De Silva, Chaudhari, and Vogelstein (2025)*

2022 **Best Short Paper Award, ECCV Workshop on Out-of-Distribution Generalization in Computer Vision**, *De Silva et al. (2022)*

2019 **Kavli NDI Distinguished Postdoctoral Fellow**, *Celine Drieu, PhD*

2019 **Kavli NDI Distinguished Postdoctoral Fellow**, *Austin Graves, PhD*

2019 **Winner of Pistritto Fellowship.**, *Vivek Gopalakrishnan*

2017 **Kavli NDI Distinguished Postdoctoral Fellow**, *Audrey Branch, PhD*

2017 **Best Presentation Award HPDC**, *Mhembere et al. (2017)*

2017 **Nonparametric Statistics of the American Statistical Association Student Paper Award**, *Lee et al. (2017)*

2014 **F1000 Prime Recommended**, *Vogelstein et al. (2014)*

2013 **Spotlight**, *Neural Information Processing Systems (NIPS)*

2011 **Trainee Abstract Award**, *Organization for Human Brain Mapping*

2008 **Spotlight**, *Computational and Systems Neuroscience (CoSyNe)*

OTHER MEDIA

2026 **"Jeff Bezos Is Funding a Wild Hunt for the Brain's 'Core Algorithm'"**, *Wired*
https://www.wired.com/story/jeff-bezos-is-funding-a-wild-hunt-for-the-brains-core-algorithm/

2026 **"Catalio's Neuroscience Startup Flourish Emerges With Funding from Bezos, Google Ventures"**, *citybiz*
https://www.citybiz.co/article/856404/catalios-neuroscience-startup-flourish-emerges-with-funding-from-bezos-google-ventures/

2025 **"New Method Advances Reliability of AI with Applications in Medical Diagnostics"**, *Johns Hopkins Medicine*
https://www.hopkinsmedicine.org/news/newsroom/news-releases/2025/08/new-method-advances-reliability-of-ai-with-applications-in-medical-diagnostics

2024 **"BI 189 Joshua Vogelstein: Connectomes and Prospective Learning"**, *Brain Inspired (podcast)*
https://braininspired.co/podcast/189/

2024 **"From Cells to Consciousness: Joshua Vogelstein on Redefining Learning in AI"**, *newpush (YouTube)*
https://www.youtube.com/watch?v=oYx0hNVx6U4

2023 **"Ancestral Instinct vs. Artificial Intelligence with Joshua Vogelstein"**, *Sacred Sons (YouTube)*
https://www.youtube.com/watch?v=1YaT4RpflO8

2023 **"Why scientists just mapped every synapse in a fly brain"**, *NPR*
https://www.npr.org/2023/03/28/1166541020/why-scientists-just-mapped-every-synapse-in-a-fly-brain

2023 **"The First Fully Mapped Animal Brain Is The Larva Of A Fruit Fly"**, *WNYC Studio*
https://www.sciencefriday.com/segments/animal-brain-map-fruit-fly

2020 **"Prazosin Might Be A Treatment for COVID-19. More Data Is Urgently Needed"**, *Forbes*
https://www.forbes.com/sites/stevensalzberg/2020/04/08/a-possible-treatment-for-covid-19/?sh=9055d4975af2

2020 **"Why Some People Get Sicker Than Others: COVID-19 is proving to be a disease of the immune system."**, *The Atlantic*
https://www.theatlantic.com/health/archive/2020/04/coronavirus-immune-response/610228/

2020 **"Coronavirus Can Set Off a 'Cytokine Storm.' These Drugs May Calm It"**, *The New York Times*
https://www.nytimes.com/2020/06/11/health/coronavirus-cytokine-storm.html

2020 **"Too Much Caution Is Killing Covid Patients"**, *The Wall Street Journal*
https://www.wsj.com/articles/too-much-caution-is-killing-covid-patients-11606238928

2020 **"Researchers Urge Clinical Trial of Blood Pressure Drug to Prevent Lethal Complication of Covid-19"**, *Johns Hopkins Medicine*
https://www.hopkinsmedicine.org/news/newsroom/news-releases/researchers-urge-clinical-trial-of-blood-pressure-drug-to-prevent-lethal-complication-of-covid-19#: :text=Researchers%20in%20the%20Ludwig%20Center,SARS%2DCoV%2D2%20infection.

2020 **"Preventing 'Cytokine Storm' May Ease Severe COVID-19 Symptoms"**, *Howard Hughes Medical Institute*
https://www.hhmi.org/news/preventing-cytokine-storm-may-ease-severe-covid-19-symptoms

2019 **"BME Pioneers: Joshua Vogelstein"**, *BME Pioneers*, https://vimeo.com/hartlove/review/331872774/9dacde8acb

2019 **"Technology Connecting the Brain to the Human Experience"**, *Tomorrows Discoveries*,
https://www.youtube.com/watch?v=NqmbDPzJBgI

2018 **"Web service makes big data available to neuroscientists"**, *Nature*, https://www.nature.com/articles/d41586-018-07195-2

2016 **"Johns Hopkins researchers want to use big data to chart the brain"**, *Johns Hopkins University*
https://hub.jhu.edu/2016/11/02/johns-hopkins-neuroscientists-brain-trust/

2016 **"International brain projects proposed"**, *Science*, http://science.sciencemag.org/content/352/6283/277

2016 **"International Brain Initiative"**, *Kavli*, http://www.kavlifoundation.org/international-brain-initiative

2016 **"Worldwide brain-mapping project sparks excitement — and concern"**, *Nature*
http://www.nature.com/news/worldwide-brain-mapping-project-sparks-excitement-and-concern-1.20658

2016 **"International Brain Projects Considered"**, *BRAIN initiative*
http://www.braininitiative.org/2016/04/22/international-brain-projects-considered/

2016 **"Three Grand Challenges for Brain Science That Can Be Solved in 10 Years"**, *MIT Technology Review*
https://www.technologyreview.com/s/602274/three-grand-challenges-for-brain-science-that-can-be-solved-in-10-years/

2016 **"International Brain Initiative Launch and VIP Dialog: Towards an International Brain Station"**, *US Department of State*
https://2009-2017.state.gov/r/pa/prs/ps/2016/09/262200.htm

2015 **"The Open Connectome Project takes a close look at the brain"**, *Johns Hopkins Magazine*
https://hub.jhu.edu/magazine/2015/winter/open-connectome-project-joshua-vogelstein-randal-burns/

2014 **"Johns Hopkins mathematician receives grant to support study of brain's circuitry"**, *Johns Hopkins University*
https://hub.jhu.edu/2014/08/22/cary-priebe-nsf-eager-grant/

2014 **"Mapping Neural Activation onto Behavior in an Entire Animal"**, *Science*
http://science.sciencemag.org/content/344/6182/372.full

2014 **"Researchers create a reference atlas for neural circuits in fruit fly larvae"**, *MedicalXpress*
https://medicalxpress.com/news/2014-03-atlas-neural-circuits-fruit-larvae.html

2014 **"Linking Neurons to Behaviors"**, *TheScientist*
http://www.the-scientist.com/?articles.view/articleNo/39571/title/Linking-Neurons-to-Behaviors

2014 **"Ten thousand neurons linked to behaviors in fly"**, *ScienceNews*, https://www.sciencenews.org/node/188288

2014 **"Fly brain 'atlas' opens door to linking human neurons to actions"**, *Reuters*
http://www.reuters.com/article/us-science-flybrain/fly-brain-atlas-opens-door-to-linking-human-neurons-to-actions-idUSBREA2Q23B20140327

2014 **"Research Highlights"**, *Nature*, http://www.cis.jhu.edu/ parky/CEP-Publications/Nature-highlight.pdf

2012 **""Open Access to the Brain" – Podcast 1: Joshua Vogelstein"**, *The Connectome Podcast*
https://theconnectome.wordpress.com/2012/02/05/podcast-1-our-interview-with-joshua-vogelstein/

2012 **"Comment on "The Predictive Capacity of Personal Genome Sequencing""**, *Science*
http://stm.sciencemag.org/content/4/135/135le3.full

2012 **"Comment on "The Predictive Capacity of Personal Genome Sequencing""**, *Science*
http://stm.sciencemag.org/content/4/135/135le5.full

2011 **"Mapping the Mind"**, *Johns Hopkins Engineering*, https://engineering.jhu.edu/magazine/2011/09/mapping-mind/#.W9s9xJNKiPo

2011 **"Q&A: What is the Open Connectome Project?"**, *BMC Nature*
https://neuralsystemsandcircuits.biomedcentral.com/articles/10.1186/2042-1001-1-16

PROFESSIONAL/SOCIAL MEDIA PRESENCE

@neuro_data, *X (Twitter) account, 7,715 followers, 4,852 posts, joined March 2011*

Bits and Brains
Professional blog reguarding all things academic, neurological, and statistical, with approximately 30 blog posts, approximately one new post per month (9,000 page views, 3,200 unique users)
Most Popular Post: 10 Simple Rules to Write a Paper from Start to Finish

medium.com/@progl, *My Medium account where I post articles on both personal and professional topics*

TRANSLATION / TECHNOLOGY TRANSFER ACTIVITIES

OPEN DATASETS

2019 – **Templier et al. (2019)**, *34 citations, 119 unique visitors*
The non-destructive collection of ultrathin sections onto silicon wafers for post-embedding staining and volumetric correlative light and electron microscopy using MagC. MagC allows the correlative visualization of neuroanatomical tracers within their ultrastructural volumetric electron microscopy context

2018 – **Bloss et al. (2018)**, *149 citations, 530 unique visitors*
Images of CA1 pyramidal neurons for analysis involving feature-selective firing as a result of dendritic integration of inputs from multiple brain regions. Show that single presynaptic axons form multiple, spatially clustered inputs onto the distal, but not proximal, dendrites of CA1 pyramidal neurons

2018 – **Branch (2018)**, *2 citations, 223 unique visitors*
Adult generated neurons in aging M. musculus imaged using array tomography, multi-spectral light microscopy, and electron microscopy

2017 – **Allen Atlas**, *142 citations, 1058 unique visitors*
Anatomical reference atlases that illustrate the adult mouse brain in coronal and sagittal planes. They are the spatial framework for datasets such as in situ hybridization, cell projection maps, and in vitro cell characterization.atlas.brain-map.org

2017 – **Hildebrand et al. (2017)**, *416 citations, 1,014 unique visitors*
A multi-resolution serial-section electron microscopy data set containing the anterior quarter of a 5.5 days post fertilization larval zebrafish, including its complete brain acquired by Hildebrand and colleagues. Electron micrographs and reconstructions are available for view in CATMAID

2017 – **Tobin et al. (2017)**, *150 citations, 43 unique visitors*
Wiring variations that enable and constrain neural computation in a sensory microcircuit

2016 – **Bloss et al. (2016)**, *189 citations, 701 unique visitors*
Images of molecularly defined inhibitory interneurons and CA1 pyramidal cell dendrites collected using correlative light-electron microscopy and large-volume array tomography

2016 – **Dyer et al. (2016)**, *100 citations, 216 unique visitors*
Mesoscale (1 cubic micron resolution) resolution images generated with the use of synchrotron X-ray microtomography (microCT) from millimeter-scale volumes of mouse brain. X-ray tomography promises rapid quantification of large brain volumes

2016 – **Lee et al. (2016)**, *590 citations, 725 unique visitors*
Electron microscopy data collected at 4 × 4 × 40 nm per voxel from the visual cortex in Mouse V1 used in a study of an excitatory network

2016 – **Wanner et al. (2016**, *45 citations, 328 unique visitors*
Serial block face scanning EM (SBEM) and conductive sample embedding image stack from an olfactory bulb (OB) of a zebrafish larva at a voxel resolution of 9.25 × 9.25 × 25 nm3

2015 – **Amunts et al. (2013)**, *948 citations, 1,041 unique visitors*
BigBrain is an ultrahigh-resolution three-dimensional model of a full human brain at 20 micrometer resolution, enabling an unprecedented look into the human brain at micro- and macro-scopic scale

2015 – **Bhatla et al. (2015)**, *83 citations, 467 unique visitors*
Nikhil Bhatla and Rita Droste in Bob Horvitz's Lab reconstruction of the anterior half of the C. elegans feeding organ, the pharynx. Volumes for three adult hermaphrodite worms include volumetric tracing of all neurons, selected cell types, I2 neuron synapses. 50 nm thick sections with an image resolution of 2 nm per pixel

2015 – **Collman et al. (2015)**, *130 citations, 382 unique visitors*
Mouse cortex collected using conjugate array tomography (AT), a volumetric imaging method that integrates immunofluorescence and EM imaging modalities in voxel-conjugate fashion

2015 – **Deisseroth et al. (2014)**, *877 citations, 208 unique visitors*
Twelve CLARITY mouse brains (5 wild type controls and 7 behaviorally challenged) were prepared by Li Ye, and imaged using CLARITY-Optimized Light-sheet Microscopy (COLM) (whole brain COLM imaging and data stitching performed by R. Tomer, in preparation)

2015 – **Harris et al. (2015)**, *55 citations, 463 unique visitors*
Three volumes of hippocampal CA1 neuropil in adult rat imaged by the laboratory of Kristen M Harris, PhD, at an XY resolution of 2 nm on serial sections of 50-60 nm thickness

2015 – **Kasthuri et al. (2015)**, *1,161 citations, 1,299 unique visitors*
Saturated reconstruction of a sub-volume of mouse neocortex collected using automated technologies in which all cellular objects (axons, dendrites, and glia) and many sub-cellular components are rendered and itemized in a database. Provides access to the complexity of the neocortex and enables further data-driven inquiries

2015 – **Micheva et al. (2015)**, *57 citations, 190 unique visitors*
Multi-channel array tomography data of the barrel cortex of an adult mouse (C57BL/6J)

2015 – **Ohyama et al. (2015)**, *590 citations, 299 unique visitors*
The side view of the approximately 7,000 neurons reconstructed so far, either in full or partially, of the approximately 12,000 neurons of the central nervous system of Drosophila larva. The 0111-8 data set was originally sectioned and imaged by Richard D. Fetter and his two tech assistants

2015 – **Randlett et al. (2015)**, *601 citations, 498 unique visitors*
Zebrafish brain atlas with surface mesh of different regions intended for the analysis of whole-brain activity mapping

2014 – **Weiler (2014)**, *15 citations, 123 unique visitors*
Images of whisker-associated barrel columns of mouse somatosensory cortex stained with antibodies against selected antigens (DAPI, YFP), and indirect immunofluorescence. Images collected by the lab of Stephen J Smith

2013 – **Bumbarger et al. (2013)**, *187 citations, 22 unique visitors*
Serial, thin section data generated by Dan Bumbarger in Ralf Sommer's lab in order to compare the pharyngeal connectomes of the pharyngeal nervous system between Caenorhabditis elegans and Pristionchus pacificus. In P. pacificus they found clearly homologous neurons for all of the 20 pharyngeal neurons in C. elegans, and massive rewiring of synaptic connectivity between the two species

2013 – **Takemura et al. (2013)**, *758 citations, 144 unique visitors*
The right part of the brain of a wild-type Oregon R female fly that was serially sectioned into 40-nm slices. A total of 1,769 sections, traversing the medulla and downstream neuropils, were imaged at a magnification of 35,000X

2011 – **Bock et al. (2011)**, *952 citations, 511 unique visitors*
Volume of mouse primary visual cortical data, spanning layers 1, 2/3, and upper layer 4 collected as electron microscope (EM) data and two-photon microscopy data collected by Davi Bock, Ph.D. and Wei-Chung Allen Lee, Ph.D.. Images have a resolution of 4x4x45 cubic nanometers

OPEN-SOURCE SOFTWARE: ACTIVE

2025 – **AutoGMM**, *4 stars, 1 fork*, Python package for automatic Gaussian mixture modeling

2021 – **treeple (formerly scikit-tree)**, *88 stars, 26 forks*
Scikit-learn-compatible package for modern tree-based algorithms for supervised and unsupervised learning problems. Extends the robust API of scikit-learn for tree algorithms that achieve strong performance in benchmark tasks.

2020 – **hyppo**, *252 stars, 96 forks*
An open-source software package for multivariate hypothesis testing. It intends to be a comprehensive multivariate hypothesis testing package that runs on all major operating systems, and includes novel tests not found in other packages.

2020 – **brainlit**, *30 stars, 17 forks*
This repository is a container of methods that Neurodata uses to expose their open-source code while it is in the process of being merged with larger scientific libraries such as scipy, scikit-image, or scikit-learn. Additionally, methods for computational neuroscience on brains too specific for a general scientific library can be found here, such as image registration software tuned specifically for large brain volumes.

2019 – **graspologic (Graph Statistics)**, *1,008 stars, 173 forks*
Co-developed with Microsoft Research: Utilities and algorithms designed for processing and analysis of graphs with specialized graph statistical algorithms

2019 – **neuroparc**, *222 stars, 52 forks*
This repository contains a number of useful parcellations, templates, masks, and transforms to (and from) MNI152NLin6 space. The files are named according to the BIDs specification

2018 – **m2g (MR graph analysis)**, *65 stars, 36 forks*
A Python pipeline which uses diffusion MRI data from individuals to generate connectomes reliably and scalably

OPEN-SOURCE SOFTWARE: CONTRIBUTED

2019 **cloud-volume**, *Added support for additional file types*

2019 **C-PAC**, *Added streamlined reproducible pipeline*

2019 **scipy**, *Added mgc, a state of the art method for hypothesis testing we developed in the lab*

2018 – 2019 **neuroglancer**, *Added multispectral support to enable light microscopy data use*

2018 **igraph**, *Added spectral clustering functionality*

2017 – 2018 **render**, *Added cloud support*

2017 **boss**, *Developed core functionality*

OPEN-SOURCE SOFTWARE: ARCHIVED

2020 – 2022 **ProgLearn (Progressive Learning)**, *39 stars, 44 forks*, A Python package for exploring and using progressive learning algorithms

2019 – 2020 **ARDENT (Affine and Regularized Deformative Numeric Transform)**, *17 stars, 6 forks*
A Python package for performing automated image registration using LDDMM

2019 – 2021 **reg (Image registration)**, *23 stars, 4 forks*
A Python package which performs non-linear affine and deformable image registration

2017 – 2024 **Sparse Projection Oblique Randomer Forests (Classification and regression)**, *105 stars, 47 forks*
SPORF is an improved random forest algorithm that achieves better accuracy and scaling than previous implementations on a standard suite of > 100 benchmark problems. Superseded by treeple, listed under Active

2019 – 2020 **Uncertainty-Forest**, *12 stars, 5 forks*
A Python package containing estimation procedures for posterior distributions, conditional entropy, and mutual information between random variables X and Y

2018 – 2019 **LOL (Supervised dimensionality reduction)**, *24 stars, 37 forks*
Linear Optimal Low-rank (LOL) projection for improved classification performance in high-dimensional classification tasks

2018 – 2021 **MGC (Non-parametric hypothesis testing)**, *33 stars, 51 forks*
Multiscale Graph Correlation (MGC) is a framework for universally consistent testing high-dimensional and non-Euclidean data. Superseded by hyppo, listed under Active

2018 – 2019 **ndcloud (NeuroData Cloud)**, *The deployment of tools which support the Open Connectome Project*

2016 – 2019 **Non-Parametric-Clustering**, *6 stars, 2 forks*
A program which uses non-parametric-clustering to minimize or maximize a given criterion function

2017 – 2019 **ndex**, *4 stars, 0 forks*
Python 3 command-line program to exchange (download/upload) image data with NeuroData's cloud deployment of APL's BOSS spatial database

2017 – 2019 **knor (Clustering)**, *2 stars, 2 forks*
Python version of knor, a highly optimized and fast library for computing k-means in parallel with accelerations for Non-Uniform Memory Access (NUMA) architectures

2017 – 2019 **SynapseAnalysis (Synapse Detection)**, *4 stars, 1 fork*
A framework to evaluate synaptic antibodies for array tomography applications

2017 – 2018 **MEDA (Matrix Exploratory Data Analysis)**, *0 stars, 1 fork*, A python package for matrix exploratory data analysis

2017 – 2018 **ndwebtools**, *0 stars, 1 fork*
ndwebtools (ndweb) is a Django application to provide a user-friendly interface for interacting with NeuroData resources and data

2015 – 2018 **ndviz**, *9 stars, 3 forks*, Web visualization and analysis tools for neuroimaging datasets, powered by Neuroglancer

2015 – 2016 **DMG**, *26 stars, 7 forks*
An implementation of a distributed multigrid Poisson solver for image stitching, smoothing, and sharpenting

2015 **VESICLE (EM Synapse Detection)**, *3 stars, 4 forks*
Reference synapse detection program for processing serial electron microscopy data

2015 **CAJAL**, *6 stars, 6 forks*
A MATLAB API that provides a simple to use interface with Open Connectome Project servers and provides RAMON Objects, unit tests, configuration scripts, and utilities

2012 – 2017 **FlashGraph (Scalable Analytics)**, *237 stars, 51 forks*
General-purpose graph analysis framework that exposes vertex-centric programming interface for users to express varieties of graph algorithms

2012 – 2017 **FlashX (Scalable machine learning)**, *237 stars, 51 forks*
A matrix computation engine that provides a small set of generalized matrix operations on sparse matrices and dense matrices to express varieties of data mining and machine learning algorithms

2011 – 2016 **oopsi (Calcium Spike Sorting)**, *24 stars, 9 forks*, Model-based spike train inference from calcium imaging

2011 – 2017 **ndstore**, *37 stars, 12 forks*
Scalable database cluster for the spatial analysis and annotation of high-throughput brain imaging data

CONSULTANCY

2017 **Consultant**, *Greenspring Associates*

2016 **Consultant**, *Scanadu*

ADVISORY BOARD APPOINTMENTS

05/26 – **Venture Partner**, *Catalio Capital Management*, *Healthcare and life sciences investment firm*

2018 – 2022 **Advisory Board**, *Mind-X*

A neurotechnology company combining brain-computer interfaces and artificial intelligence to make the world's information available with the speed and ease of a single thought.

Acquired by Blackrock Neurotech, April 2022, 15 employees.

2017 – **Advisory Board**, *PivotalPath*

PivotalPath is a leading hedge fund research and intelligence organization built by a team of experienced alternative investment professionals and fintech developers.

Raised undisclosed amount of funding, 11 employees.

STARTUPS

12/25 – **Co-Founder**, *Flourish*

Neuroscience-grounded AI company building brain-inspired, low-power, continually-learning intelligence. Backed by Jeff Bezos, Lux Capital, Google Ventures, Catalio Capital, and ARCH Venture Partners

Raised $500M; valued at approximately $2.5B

2017 – 2022 **Co-Founder**, *gigantum*

The future of data science is open, decentralized and user friendly. That is why we created a platform that enables anybody to create and share totally reproducible computational work with the world.

Acquired by NVIDIA in early 2022, 15 employees.

2012 – 2019 **Chief Technical Officer**, *global domain partners*

Global Domain Partners is a quantitative hedge fund, founded in 2005, that was acquired by Mosaic Investment Partners in 2012

6 employees.